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Article

Multi-step stability selects, actuator-term survival explains: physics-informed features have a non-monotone effect on sparse surrogate models for economic greenhouse climate control

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Abstract: Economic model predictive control (MPC) of greenhouse climate needs a process model, and sparse identification of nonlinear dynamics offers an interpretable surrogate; which open-loop criterion should select one is rarely tested. We screened 72 identification configurations on the GreenLight tomato model under ERA5 weather and compared fifteen controllers over four test seasons at up to 20 replicates, scored by seasonal margin. One-step accuracy and multi-step stability ranked the candidate libraries in opposite orders: physically motivated features lowered one-step RMSE from 1.86 to 1.68 °C while raising median 24 h rollout RMSE from 2.67 to 24.27 °C. Multi-step stability selected correctly—the raw library also gave the best economics, $+4.32\ \mathrm{EUR\,m^{-2}}$ against $+3.09$ for the best physics-informed variant (Holm-corrected $p = 4.6 \times 10^{-4}$, family of 15) and $+2.26$ for an agronomic heuristic re-tuned under the same 16-trial budget the reinforcement-learning agents received. Conditioning does not explain this: changing only the library, the condition number rose monotonically (8.2, 24.5, 53.4) but margin did not ($+4.32$, $+0.28$, $+2.75$). Margin followed survival of the boiler coefficient under thresholding (55, 15, 55 %); reinstating it is significant but small (median $+0.21$, 16 of 20, $p = 5.9 \times 10^{-4}$). The advantage is not Pareto-dominant: five controllers hold the margin–violation front.

Keywords: greenhouse climate control; economic model predictive control; sparse identification of nonlinear dynamics; control-oriented model selection; multi-step prediction error; physics-informed feature libraries; sparsity thresholding; reinforcement learning baselines

1. Introduction

Greenhouse vegetable production is energy-intensive, and its economic result depends directly on the quality of climate control. Heating, CO_2 enrichment and supplementary lighting dominate variable cost, and each has to be weighed against the revenue from the additional yield it buys [1–2]. This is the natural setting for model predictive control (MPC): an economic criterion is optimised over a finite horizon subject to the temperature, humidity and CO_2 bounds that define a productive microclimate [3–6]. MPC, however, requires a process model, and building and calibrating a detailed physical greenhouse model is laborious, which motivates surrogate models identified from data. Sparse identification of nonlinear dynamics (SINDy) is one such route: the dynamics are recovered explicitly as a sparse combination of library functions [7–10]. Unlike a neural surrogate, a SINDy model is a white box: its equations can be written out, checked for physical and dimensional consistency, and embedded analytically in the MPC solver — which matters when a controller is to be trusted with a crop.

Whether an interpretable surrogate costs anything in control quality is an empirical question, and answering it objectively is obstructed by three defects that recur in the greenhouse-control and data-driven-control literature. First, learned controllers are often compared against a weak heuristic, whereas well-tuned agronomic rules are a strong reference [11–13]. Second, the reported performance index is frequently a setpoint-tracking error rather than economic margin [6,11], although the two need not be monotonically related. Third — and least obvious — the hyper-parameters of the surrogate are selected on open-loop prediction accuracy [9,14], tacitly assuming that a more accurate model yields better control. Table 1 states each defect, the remedy adopted here, and what we measured the defect to be worth. We emphasise that the first two defects were present in our *own* earlier analysis of this dataset — an untuned reference, and an MPC stage cost whose actuator weights contradicted the prices the economic criterion itself uses — and that correcting them removed about two thirds of the effect we had originally reported.

The third assumption is exactly what the identification-for-control literature warns against: a model should be judged by the behaviour of the loop it closes, not by the quality of its open-loop fit [15–17]. The same phenomenon reappears for learned models under the name *objective mismatch*: one-step predictive accuracy is a poor proxy for control performance [18–20], and recent surveys confirm that the point is live for modern data-driven surrogates [21–22]. That literature is largely diagnostic, and its prescription — evaluate candidates in closed loop — costs a simulator, a solver and a full season per candidate. The practical question is therefore narrower and, to our knowledge, unanswered for economic greenhouse MPC: *if* a surrogate must be selected open-loop, *which* open-loop criterion tracks closed-loop economics?

A second, related gap concerns the content of the feature library. Physically motivated basis functions — saturation vapour pressure, vapour-pressure deficit, screen-attenuated radiation — are adopted almost by default in physics-informed surrogate modelling [23–24], on the reasonable presumption that encoding known physics can only help. They are rarely tested for harm. Yet such features are nonlinear transforms of the same measured states, hence near-collinear with them, and collinearity is known to degrade sparse regression [25–26]. Whether that presumption survives a closed-loop economic criterion is the second question we address.

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Table 1. The three benchmarking defects addressed in this study, and the measured cost of leaving each in place. All figures are seasonal margin in EUR m^{-2} unless stated otherwise.

Defect	Remedy adopted here	Measured consequence of the defect
Weak reference: learned controllers compared with a hard-coded heuristic	The agronomic rule-based reference is tuned on the training seasons only, by a 16-trial random search — the same trial budget the reinforcement-learning agents receive	Tuning lifts the heuristic from -1.23 to $+2.26$ (gain $+3.49$), so every advantage measured against the untuned reference is overstated by that amount. The tuned heuristic buys the gain with constraint pressure (2813 to 4339 violation steps) and is itself off the Pareto front
Proxy objective: tracking error reported instead of margin, and an MPC stage cost whose actuator weights contradict the prices the criterion uses	Every controller is scored by the simulator's own profit function, jointly with constraint violations under Pareto dominance; the MPC stage cost is re-weighted to match the price vector	Re-weighting improves both axes for all eight MPC controllers: $+1.54$ to $+2.74$ for the five physics-library variants and $+1.57$ for the neural surrogate, against $+0.21$ to $+0.25$ for the two raw-library ones, cutting the apparent library effect from $+3.66$ to $+1.23$
Open-loop selection: identification configuration chosen by one-step prediction accuracy	Candidates are scored by multi-step rollout stability and regression conditioning, fixed before and independently of any closed-loop evaluation	Among the candidates that are embeddable in the solver, the best one-step configuration comes from the library with condition number 53.4 rather than the one with 8.2, and with median 24-h rollout RMSE 24.27 °C rather than 2.67 °C

We address both questions on a validated GreenLight tomato model [1,13], driven by recorded weather for a single location and simulated at a 15-min control step. Candidate surrogates were identified on two training seasons, and fifteen controllers were evaluated — nine SINDy-MPC variants, a neural-network MPC, two deep reinforcement-learning agents [27–29], a full-simulator-model MPC and the rule-based heuristic in both its stock and tuned configurations — over four held-out test seasons. A *season* here is the 60-day window from 1 March, and margin is profit per square metre of floor area. Each learned controller was run at 20 identification replicates per season (10 for the neural-network MPC); the heuristic is deterministic and was run once per season, and the full-simulator-model MPC returned only three of the four seasons, its 2022 runs having exhausted the solver-failure budget. Comparisons use non-parametric signed-rank tests with Holm correction [30–32]; against the deterministic heuristic the test is one-sample against a constant rather than genuinely paired. The contributions are four.

(i) One-step accuracy and multi-step stability rank candidate libraries in opposite orders, and the closed-loop ordering follows the multi-step one. The identification

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ladder evaluated 72 labelled configurations, covering 54 distinct fits — two of the optimizer labels turn out to be bit-identical. Within its degree-1 block — the degree-2 candidates are all non-embeddable in the MPC solver — the library with the best one-step accuracy is the worst on every multi-step metric. Taking the undenoised cells under the two sparse estimators ($n = 40$ fits per library), adding physically motivated features improves one-step RMSE from 1.86 to 1.68 °C while degrading the median 24-h rollout RMSE from 2.67 to 24.27 °C, raising the regression condition number from 8.2 to 53.4 and the divergence rate from 0.000 to 0.076 (Figure 1); withholding only the bilinear cross terms recovers part of the loss but not the ordering (rollout 10.58 °C, $\kappa = 24.5$). In closed loop, a raw, physics-free library — also the sparsest of the three — attains the best four-season mean margin of the fifteen controllers, +4.32 EUR m^{-2} against +3.09 for the best physics-informed variant, ahead in 62 of 80 season × replicate pairs ($p_{\mathrm{Holm}} = 4.6 \times 10^{-4}$, family of 15). The gap is small against the run-to-run spread — standard deviations 4.08 and 2.71 for a difference of 1.23 — and it reverses in one of the four seasons (−1.56 against +0.27 in 2021). The library ordering holds again on four seasons never used at any stage of model or controller development (four controllers, eight replicates, default objective). Physical motivation of a basis function is therefore not evidence that it belongs in the library.

(ii) The damage done by physically motivated features is not monotone in how many are added. All three libraries have now been carried into closed loop under a strict one-factor design — same estimator, same sparsity threshold, same seeds, same priced objective, four seasons × 20 replicates, zero truncated runs, the library the only thing that changes. Open-loop degradation is monotone in the number of physics features: $\kappa = 8.2 \to 24.5 \to 53.4$ and median rollout $2.67 \to 10.58 \to 24.27$ °C. Closed-loop margin is not: $+4.32 \to +0.28 \to +2.75$ EUR m^{-2}. The worst-conditioned library beats the middle one by +2.47 on average (median +2.39, 63 of 80 paired runs, $p = 1.3 \times 10^{-8}$), in each of the four seasons taken separately. Under the second sparse estimator only the two endpoints exist as closed-loop controllers (+3.83 raw, +2.48 full physics), so the raw-over-physics direction replicates there but the three-point non-monotonicity rests on one estimator. Conditioning therefore predicts open-loop multi-step stability cleanly and closed-loop economics not at all; the agreement observed while only two of the three libraries had closed-loop controllers was a two-point coincidence. This retracts an ill-conditioning mechanism we had previously advanced for the closed-loop result, and replaces it with contribution (iv).

(iii) A benchmark whose comparison defects have been measured rather than assumed. Against the properly tuned heuristic the raw-library controller is better by +2.06 EUR m^{-2}, ahead in 75 of 80 season × replicate comparisons ($p_{\mathrm{Holm}} = 6.3 \times 10^{-11}$, Holm correction over a family of 15, which we declare wherever a corrected level is quoted). Three further SINDy-MPC variants also clear the tuned heuristic significantly, by smaller margins (+1.57, +0.83, +0.82), and the two full-physics controllers are ahead of it in point estimate without being separable from it: the raw library is first, not alone. Against the untuned reference the same advantage would read +5.54 — the failure mode this literature is criticised for, reproduced and quantified on our own data. Likewise, two thirds of what initially looked like a feature-library effect was in fact a defective MPC stage cost (Table 1, row 2).

(iv) What the closed-loop outcome does track is survival of the actuator pathway through the sparsity threshold. Across the one-factor triple, the direct boiler coefficient $\xi_{u\mathrm{Boil}}$ is retained in 11 of 20 identification replicates with the raw library, 11 of 20 with full physics and 3 of 20 with physics-without-cross-terms — 55 %, 55 %, 15 %, which is the margin ordering and not the κ ordering. A surrogate whose boiler column has been thresholded away cannot be steered with the boiler, however accurate it is one step ahead

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and however well conditioned the fit that produced it. The controlled evidence is a single-coefficient ablation, the only randomised manipulation in the mechanism block: reinstating the deleted coefficient in the frozen recipe, with the library, the scalers, every other term and the MPC construction held fixed, is positive in 16 of 20 replicates ($p_{\mathrm{Holm}} = 1.2 \times 10^{-3}$ over the two knock tests) with median $+0.21$ EUR m^{-2} and mean $+1.92$ — significant, typically small and heavily right-skewed — while the complementary knock-out is null. We are explicit about the limits of this account. Survival *gates* the outcome but does not *order* it: among the two libraries that keep the term the raw one still wins by $+1.56$ EUR m^{-2}, and raw-library runs that lost the term still average $+3.48$, above physics-library runs that kept it ($+2.36$). The obvious structural account of why the middle library is the worst — that it carries enough collinearity for the threshold to destabilise the direct boiler term but, unlike full physics, has no bilinear $T_{\mathrm{in}}u_{\mathrm{Boil}}$ term to carry the heating pathway once the direct one is cut — has been tested and rejected: deleting exactly that one feature from the full library leaves conditioning ($\kappa = 52.3$), open-loop instability and closed-loop margin ($+2.11$ EUR m^{-2}, $p_{\mathrm{Holm}} = 4.4 \times 10^{-6}$ above the middle library) at the full library's level, with survival at 40 % — nowhere near the collapse a bilinear bypass predicts (Sections 3.5 and 4). Why one library loses the term is thus unresolved; that survival gates the outcome is not.

What this does not show is equally explicit. The advantage is not Pareto-dominant: the front holds five points, and the PPO agent attains far fewer constraint violations (1330 against 4130 steps per season) at a lower margin. The surrogate carries no crop state, so the MPC objective is necessarily a proxy and terminal biomass cannot be priced. The mechanism experiments are single-season and replication is unequal across controllers. The non-monotonicity rests on the middle library, which reaches closed loop under one estimator only, and survival is measured rather than manipulated at library level — the ablation is the only randomised contrast anywhere in the mechanism, and the term-deletion test that removed the detour explanation is, like the library comparison itself, a controlled structural comparison rather than a randomised one. All results are simulation-only, for one location and one planting date, and reproducibility is established within one computing environment only — the cross-environment case was never measured, and the closed-loop loop is chaotic enough that even the deterministic reference does not reproduce exactly between two harnesses. The remainder of the paper is organised as follows. Section 2 describes the controlled object, the economic criterion, the data, the controllers and the identification protocol; Section 3 reports the model-selection ladder, the four-season economic comparison and the mechanism experiments; Section 4 discusses the implications and the limits; Section 5 concludes.

2. Materials and Methods

2.1. Simulator, site, weather and season

All experiments were run in simulation with the GreenLight tomato greenhouse model [1–2] through the GreenLight-Gym interface [13]. The simulated house is at Rostov-on-Don (47.24° N, 39.71° E). Weather forcing came from the ERA5/ERA5-Land reanalysis [33] via Open-Meteo [34]; sky temperature was derived from the retrieved dew point and cloud cover, and the deep-soil boundary condition was refitted as a single annual harmonic to ERA5-Land soil temperature at this site, replacing the simulator's Dutch default. Outdoor CO_2 was held at 400 ppm, a limitation of the model. Each season spans 60 days from 1 March at a 15-min control step ($T_s = 900$ s), i.e. 5760 control steps per season; that count is the truncation gate used throughout.

Years were separated to prevent leakage: identification and every selection decision used 2018–2019 only, closed-loop evaluation used 2020–2023 (2020 in-distribution, 2021–

2023 out-of-distribution), and a further block of four seasons, 2014–2017, was generated after all recipes had been frozen and entered no stage of training, selection or tuning. Training trajectories were collected by the rule-based controller with a pseudo-random binary excitation of amplitude 0.3 on its actions plus an additive perturbation of scale 0.1; the condition number of the regression matrix was recorded for every fit [25]. Evaluation rollouts carry no excitation.

2.2. Economic criterion

Controllers are scored by the profit term of the simulator's own reward function; no tracking-error proxy is used. At step k that term is a marginal income π_k, and the seasonal economic performance indicator (EPI), written J throughout, is

$$J = \sum_k \pi_k = \sum_k (g_k - c_k), \tag{1}$$

where g_k is income from modelled fruit dry-matter growth, converted to fresh mass with a dry-to-fresh ratio of 0.065 and valued at 1.60 EUR/kg, and c_k are the variable costs of heating (0.09 EUR/(kW·h)), supplementary lighting (0.30 EUR/(kW·h)) and CO_2 enrichment (0.30 EUR/kg). Dividing the logged cost columns by the logged energy and CO_2 columns of regen/results/final/main.csv returns exactly these three prices across all 780 retained runs. J is *not* the scalar the environment returns as its reward: that scalar rescales the profit and subtracts normalised corridor-violation penalties. The distinction matters, because the two reinforcement-learning controllers are trained on the penalised reward while every controller is scored on the unpenalised J — their position on the violation axis partly reflects optimising a different objective from the one reported.

What J excludes matters for every result that follows. It omits planting material, labour, water, fertiliser, depreciation and fixed costs; it uses modelled dry-matter growth rather than realised marketable yield; and, critically, it does not value the biomass standing in the house at the end of the 60-day window. The surrogates of Section 2.4 carry a three-component climate state and no crop state, so terminal biomass cannot be priced inside the optimisation even in principle. That is why the MPC stage cost (Section 2.6) is a proxy rather than a discretisation of Equation (1), and why J is reported as a *margin* and not as a farm profit: a controller that harvests within the window is rewarded over one that builds a crop for a later window, and no result here separates the two.

J carries no penalty for leaving the productive range, so margin can be bought with violations. The simulator's productive band is $CO_2 \in [300,1600]$ ppm, air temperature $[15,34]$ °C and relative humidity $[50,85]$ %. Controllers are therefore compared by Pareto dominance in two axes — J and the seasonal count of steps in violation — and no scalarised "margin-minus-penalty" criterion is introduced, because the simulator does not express violations in money. This corridor is also not the one the predictive controllers minimise against: their stage cost (Section 2.6) penalises humidity only *above* 85 % and CO_2 only *below* a floor, so the lower humidity and upper CO_2 bounds are unweighted in the optimisation yet counted on the violation axis.

2.3. Controllers compared

Fifteen controllers were compared (Table 2). Two naming corrections apply relative to earlier reports of this work, and both are stated in the text because each was previously used to support a claim it does not support. First, the *low-threshold* SINDy variant is not a reduced first-principles grey box: it is the same sparse estimator on the same feature library at a threshold of 10^{-6}, i.e. the least sparse member of the family. Second, the aggregation loop applied to two SINDy controllers rolls the learner's own MPC in closed loop on the training seasons, appends the visited states to the identification set and re-estimates the model (3 iterations of 5-day episodes). The rule-based controller is *never queried* in that loop: this is on-policy re-identification, not DAgger and not imitation learning, and the rule-based controller is not an expert in it. One asymmetry inside that loop survives into the headline comparison and is disclosed here: its episodes are rolled under the default stage cost, so under the priced objective these two controllers aggregate their training states under one cost function and are evaluated under another.

Table 2. Controllers compared. Identification recipes are read from the frozen configuration regen/regen_config.py; "library" refers to Section 2.4. All predictive controllers share one horizon ($N = 20$) and one solver-failure budget (100).

Label	Family	Definition
rule_based	heuristic	agronomic setpoint heuristic, hard-coded setpoints; deterministic
rule_based_tuned	heuristic	the same heuristic after a 16-trial setpoint search (Section 2.7)
sindy_mpc_raw	SINDy-MPC	raw library, STLSQ, $\lambda = 0.05$
sindy_mpc_raw_ens	SINDy-MPC	raw library, ensemble, $\lambda = 0.05$
sindy_mpc_conf	SINDy-MPC	frozen recipe: physics_no_cross, ensemble, $\lambda = 0.05$
sindy_mpc_phys	SINDy-MPC	physics (full, with cross terms), STLSQ, $\lambda = 0.05$
sindy_mpc_phys_ens	SINDy-MPC	physics (full, with cross terms), ensemble, $\lambda = 0.05$
sindy_mpc_dense	SINDy-MPC	physics_no_cross, STLSQ, $\lambda = 10^{-3}$
sindy_mpc_lowthr	SINDy-MPC	physics_no_cross, STLSQ, $\lambda = 10^{-6}$ (not a grey box)
sindy_mpc_conf_dagger	SINDy-MPC	conf plus on-policy re-identification
sindy_mpc_dense_dagger	SINDy-MPC	dense plus on-policy re-identification
nn_mpc	neural MPC	64×64 MLP surrogate on the physics library, 300 epochs; same horizon and stage cost, but the network is not embeddable in the NLP, so its MPC is a shooting formulation solved by L-BFGS-B on autograd gradients (25 iterations per step)

Label	Family	Definition
ppo, sac	reinforcement learning	PPO [27] and SAC [28] from Stable-Baselines3 [29], $2 \cdot 10^5$ environment steps
oracle_mpc	idealised	cross-entropy planning directly on the simulator integrator: 48 candidate sequences, 2 refinement iterations, elite fraction 0.2

Four of the SINDy-MPC labels form a deliberate one-factor design over the feature library, and the mechanism analysis of Section 3.5 rests on it. Holding the polynomial degree (1), the denoiser (none), the estimator and the sparsity threshold (0.05) at the frozen recipe's values, and changing *only* Θ, gives sindy_mpc_raw_ens (raw), sindy_mpc_conf (physics_no_cross) and sindy_mpc_phys_ens (physics) under the ensemble estimator, and sindy_mpc_raw and sindy_mpc_phys under STLSQ. The ensemble arm is a complete three-point series; the STLSQ arm has only the two endpoints, because the two physics_no_cross STLSQ controllers were frozen at thresholds 10^{-3} and 10^{-6} rather than 0.05. Any statement about how the library ordering behaves *between* the endpoints therefore rests on the ensemble arm alone, and is scoped that way throughout.

The 16-trial budget given to the rule-based reference (Section 2.7) matches the budget the reinforcement-learning controllers are recorded as having received in the earlier pipeline, so the reference is not an unsearched baseline. That match is a stated budget, not a measured one: here the agents are trained once per seed at stock Stable-Baselines3 defaults and the neural surrogate at a fixed architecture, and no hyper-parameter search artefact for either exists in the results tree. The prediction horizon was *not* tuned for any predictive controller either; that asymmetry runs against the MPC family and is disclosed in the Discussion.

2.4. Feature libraries and sparse identification

Sparse regression [7–9] recovers an explicit discrete one-step map

$$x_{k+1} = \Xi\, \Theta(x_k, u_k, d_k), \tag{2}$$

with state $x_k = (T_{\mathrm{in}}, c, \varphi)^\top$ (air temperature, CO_2, relative humidity), six controls u_k (boiler, CO_2 injection, thermal screen, ventilation, lamps, blackout screen) normalised to [0,1], and disturbances d_k (outdoor temperature, global radiation, outdoor CO_2, two diurnal harmonics). Three feature libraries were compared, differing only in Θ:

- raw — 11 features: the five disturbance channels and the six controls, with no state transform whatsoever;
- physics_no_cross — those 11 plus three physically motivated transforms of state, radiation and screen position (14 features): saturation vapour pressure $e_s(T_{\mathrm{in}}) = 0.6108 \exp(17.27\, T_{\mathrm{in}}/(T_{\mathrm{in}} + 237.3))$ by the Magnus–Tetens relation, vapour-pressure deficit $D = (1 - \varphi/100)\, e_s$, and screen-attenuated radiation $S_{\mathrm{eff}} = I\,(1 - u_{\mathrm{thscr}})$;
- physics — those 14 plus four bilinear state × control cross terms ($T_{\mathrm{in}} S_{\mathrm{eff}}$, $\varphi u_{\mathrm{vent}}$, $(c - c_{\mathrm{out}}) u_{\mathrm{vent}}$, $T_{\mathrm{in}} u_{\mathrm{boil}}$), 18 features.

The three physical functions are nonlinear transforms of variables already in the raw library: they add no new measurement, only near-collinear columns. States and features are standardised before thresholding, so the threshold acts on dimensionless quantities, and the coefficients are transformed back to physical units. Four estimators were available: STLSQ, SR3, a constrained variant, and an ensemble optimiser with 20 bagged bootstrap replicates [10]. Because the ensemble re-draws its bootstrap, a single fit is one realisation of a lottery; the draw is therefore carried as an explicit design axis, and the number of draws is recorded per row rather than assumed — 10 per seed in the canonical wave, 6 per seed in the later wave that added the raw-library controller.

2.5. Identification ladder and pre-registered open-loop criteria

The identification recipe was selected in an open-loop ladder computed on the training years alone: 3 libraries × 2 polynomial degrees × 4 estimators × 3 denoisers = 72 configuration labels, each fitted on 20 seeds, for 1440 fits (regen/results/ladder_rerun/). No closed-loop quantity enters it. Each fit records the one-step RMSE, the multi-step rollout RMSE and divergence fraction at horizons of 4, 20 and 96 steps (1 h, 5 h, 24 h; the reported value is always the 24-h figure), the number of non-zero terms, the regression condition number, and whether the map can be embedded analytically in the MPC. The pre-registration declared two gates, MPC-embeddability and a sign-and-dimension transparency check, and specified multi-step stability only qualitatively, as a divergence fraction near zero. What was applied differs on both counts and is reported as applied: the divergence criterion was operationalised as a fixed threshold of 0.05 in the table generator, and the transparency column is empty in all 1440 rows, so that gate was never evaluated and is not claimed here. Under embeddability and the 0.05 threshold, 32 of the 72 labels pass. Figure 1a shows the degree-1 undenoised block of this grid in the plane of the two prediction criteria, with the 0.05 gate encoded by marker fill.

Four properties of this grid bound what it can support. (i) The rollout metrics are computed on the same 2018–2019 data the model was fitted to and are therefore in-sample; a separate held-out block (240 fits: 10 seeds × 3 libraries × 2 estimators × 4 fit-year → eval-year directions, regen/results/holdout/) splits the two training years to test whether the library ordering survives. (ii) The constrained and sr3 estimators produce bit-identical fits in all 360 seed-cells, so the ladder holds 54 distinct configurations under 72 labels and pooled per-library averages double-weight that estimator. (iii) Every degree-2 configuration is non-embeddable, so the embeddability gate is exactly the degree-1 filter here and carries no independent information. (iv) An earlier harness passed the rollout horizons in days rather than steps, inflating the divergence fraction of divergence-prone configurations; only the recomputed ladder is used here.

2.6. MPC formulation and stage cost

The explicit form of Equation (2) allows it to be embedded analytically in a finite-horizon optimal control problem [4]:

$$\min_{u_0,\dots,u_{N-1}} \sum_{k=0}^{N-1} [\ell_{\mathrm{res}}(u_k) + \ell_{\mathrm{corr}}(x_k)] + \ell_{\mathrm{corr}}(x_N) \quad \mathrm{s.t.} \quad x_{k+1} = \Xi\,\Theta(x_k, u_k, d_k), \tag{3}$$

with $N = 20$ steps (5 h). The corridor term is $\ell_{\mathrm{corr}} = 30\,b_T^2 + 3 \cdot 10^{-4} b_c^2 + 8\,b_\varphi^2$, where b_T, b_c and b_φ are excursions from a diurnally varying productive band: temperature in $[15 + 3w,\, 19 + 5w]$ °C, CO_2 above $400 + 300w$ ppm and relative humidity below 85 %, with $w \in [0,1]$ a smooth day weight. Violations are penalised softly so the problem stays feasible; temperature is hard-bounded to $[12,35]$ °C, ventilation to $[0,0.4]$ and the other controls to $[0,1]$, and a move-suppression term penalises control increments. The problem is solved by an interior-point method (IPOPT [35] through do-mpc [36]), whose initial guess is set once from the measured initial state and carried forward by the solver itself; no controller-specific warm start is supplied. The budget is 100 solver failures, shared by every solver-based controller. When a solve fails the step falls back to a fixed action (boiler 0.3, thermal screen closed, all else zero) and the failure counts against that budget; a run that exhausts it is excluded (Section 2.8).

The resource term ℓ_{res} of Equation (3) required correction, and this is a methodological change rather than a tuning choice. Its default weights, $20u_{\mathrm{boil}} + 10u_{\mathrm{lamp}} + 2u_{\mathrm{CO_2}}$, penalise the boiler twice as hard as the lamps. Dividing the logged cost columns by the logged actuator sums in regen/results/final/main.csv gives a marginal cost per unit of accumulated normalised control of 0.00293, 0.00870 and 0.00135 EUR respectively — ratios $1{:}\,2.97{:}\,0.46$ against the default $1{:}\,0.5{:}\,0.1$, so the most expensive actuator was the least penalised. A priced variant restores the measured ratios, $7.22u_{\mathrm{boil}} + 21.44u_{\mathrm{lamp}} + 3.33u_{\mathrm{CO_2}}$, scaled so the total at $u = (1,1,1)$ is 31.99 against the default 32.0. The corridor term is untouched, so the *only* factor that changes is the relative weighting of the actuators, not the energy/corridor balance. The correction is bounded by the harness, and the bound is stated rather than left to be inferred: the objective switch reaches only the main closed-loop comparison and the mechanism block. Every supporting block — unseen seasons, the shortened horizon, the bootstrap-draw axis, the design and sensitivity sweeps, adaptation, the safety guard, fault injection — calls its rollouts with the default stage cost hard-coded, so those results are default-objective whatever directory they were written to, and the coefficient- and threshold-sensitivity analyses have never been measured under the priced weights.

2.7. Rule-based reference and its tuning

Reporting a reference as "tuned" when it was not is exactly the weak-baseline defect this paper criticises in the literature, so the heuristic was given a 16-trial search budget (Table 3). Sixteen configurations were drawn i.i.d. uniformly from a fixed generator (default_rng(20260810)) and scored alongside the hard-coded configuration, which enters the search as trial 0, giving 17 evaluated configurations. Selection used the mean of J over the training seasons 2018–2019 only; the test seasons play no part in it. The heuristic is deterministic, so one run per configuration and season suffices and there is no run-to-run dispersion.

Table 3. Rule-based setpoint search. Ranges, hard-coded values and the selected configuration. Source: regen/results/n2_tune/tune_rb_n2.csv.

Setpoint	Range searched	Hard-coded	Selected
temp_setpoint_day (°C)	17.0–23.0	19.5	22.92
temp_set-point_night (°C)	14.0–20.0	16.5	14.38
co2_day (ppm)	400–1200	800	828.26
lamp_rad_sum_limit	5.0–25.0	10	7.62
Training J (mean 2018–2019, EUR/m^2)	—	-1.937	$+0.920$
Test J (mean 2020–2023, EUR/m^2)	—	-1.226	$+2.261$

Tuning is worth $+3.49$ EUR/m^2 over the four test seasons, concentrated in the two adverse ones, and moves the heuristic towards a wider diurnal swing (warmer by day, cooler by night) with a lower lamp threshold; it is bought with constraint pressure, the seasonal violation count rising from 2813 to 4339 steps. Only 2 of the 16 draws beat the hard-coded setpoints, placing the stock configuration near the 88th percentile of the searched space: it was not arbitrary, merely unsearched.

2.8. Statistical treatment

Dispersion is generated by re-collecting the identification data in every repeat, so each seed identifies a different surrogate; for the reinforcement-learning controllers the seed also re-runs training. In the main closed-loop comparison twenty seeds were run for every surrogate-based and learned controller; the supporting blocks use fewer, and their n is stated with each. Two exceptions are quoted wherever they appear: the neural MPC has 20 seeds under the default objective but only 10 under the priced one, and the idealised controller loses one whole season to the exclusion rule below. The heuristic is deterministic: the stock version was nevertheless repeated on all 20 seeds in the main wave, returning 20 identical values per season, while the tuning wave ran the stock and tuned versions once per season, so the tuned reference has $n = 4$ and no seed axis at all.

Three accounting rules precede any average. First, several waves were resumed after interruptions and their files overlap, so rows are deduplicated on (method, seed, test_year) — or on the block-specific factor key — before aggregation; the deduplicated priced pool contains 600 rows. Second, a run is excluded when it was truncated *and* exhausted the solver-failure budget, which removes 20 of the 800 rows of the canonical main table, all oracle_mpc in 2022. Third, seasons terminated by the simulator rather than the solver are retained as outcomes (42 such seasons in the canonical table), and every headline figure is recomputed three ways — as-is, over complete seasons only, and normalised by the realised season fraction — with any rank change reported.

Paired comparisons use the Wilcoxon signed-rank test [30] with Holm correction [31]. Because Holm's adjustment depends on the family, the family size is stated with every adjusted p-value, and adjusted values from different families are not comparable. One departure is fundamental rather than presentational: because both rule-based references are deterministic, comparisons against either are **one-sample signed-rank tests of the differences against a constant**, not genuinely paired tests, and the reference carries no sampling uncertainty. The same limit binds any test that treats a deterministic controller as the sample: at $n = 4$ the minimum attainable two-sided p is 0.125, so such a comparison can never reach significance whatever the effect size, and its point estimate is reported without a significance claim. An across-season spread quoted for a deterministic controller is labelled as such and is not an error bar.

Effect sizes accompany every p-value, with medians preferred to means wherever the distribution is skewed, and bootstrap confidence intervals [32,37] for aggregate quantities. This is deliberate: the data contain a contrast significant at $p = 0.012$ for a difference of 0.0066 EUR/m^2, and another whose mean is nine times its median.

2.9. Reproducibility, provenance and its limits

Random number generation is pinned per run coordinate: one helper derives the key from (experiment, controller, seed) and seeds Python, NumPy and torch through a single implementation, BLAS thread counts are fixed, and the 20 seeds are declared in the frozen configuration rather than passed on the command line. Every result row carries a config_hash (637c6b535a9e for all waves reported here) and a git_sha. The hash is identical across the default- and priced-objective waves and so cannot by itself discriminate between them, and the per-row variant fields only partly can: horizon and max_solver_failures are written by the main runner but by none of the supporting blocks, and objective only by waves generated after that flag was added — it is absent from the canonical main table, which is default-objective by construction. The identifying key is therefore the results tree and block together with these fields where present; git_sha is unknown in a few early resumed files.

A self-test runs the pipeline twice under identical seeds and compares SHA-256 digests of the training states and actions, the ensemble coefficients, the STLSQ coefficients, the neural-network weights and the closed-loop trajectory, together with the closed-loop J to twelve decimal places: all seven digests were identical on the pinned stack, and all nine with the reinforcement-learning policies added. Its coverage is narrower than the study: the closed-loop leg is a single surrogate-MPC rollout of one recipe on one season under the default objective, and no supporting block, neither the identification ladder nor the replay block, is exercised.

One caveat qualifies every reproducibility statement here, and it is a limit on what was measured rather than a measurement. The self-test establishes bit-identity *within* one environment. Nothing in the results tree establishes it *across* environments, and no such claim is made: the study was executed in a single environment, so the cross-environment comparison was never run and its magnitude is unknown. The provenance record makes the gap explicit rather than hiding it: although repro.py can emit an environment fingerprint (--fingerprint), the field is absent from all twenty run manifests under regen/results and the generated provenance map prints env_hash: n/a. Waves are separated only by git_sha — eleven distinct values across those twenty manifests at one common config_hash — which identifies the code but not the machine it ran on.

What the tree does show is that closed-loop margins move at a magnitude the identifying fields do not capture, even for a controller with no stochastic element. The hard-coded heuristic returns a four-season mean of -1.2061 EUR/m^2 in the main harness and -1.2264 in the tuning harness at identical config_hash, the per-season discrepancy growing monotonically over the four seasons (0.003, 0.013, 0.027, 0.038 EUR/m^2 for 2020–2023). A deterministic controller re-run on the same season should return the same number; that it does not is chaotic amplification of last-bit floating-point differences through a 5760-step nonlinear loop, which is the same mechanism that makes agreement across operating environments implausible *a priori*. The rule adopted here is therefore that a whole wave is regenerated in one environment and is never mixed cell-by-cell with another for closed-loop numbers. That the environment itself is not recorded in the manifests is a defect of this provenance scheme, stated here so that it is not mistaken for a cross-platform guarantee: the reproducibility claim is within-environment only.

3. Results

Throughout, the economic performance indicator (EPI) is the seasonal margin in EUR m^{-2}, and constraint pressure is the seasonal count of simulation steps outside the productive envelope. Result rows were deduplicated on (method, seed, test_year) before averaging, because several waves were resumed after interruptions and the files overlap; runs in which the solver exhausted its failure budget were excluded. All waves reported here carry the same config_hash (637c6b535a9e), which therefore identifies the code base but *not* the experimental variant; variants are identified by the per-row objective and horizon fields and by the results tree. Two objectives appear: the original stage cost, and the *priced* stage cost, in which the actuator weights were made consistent with the prices the economic criterion itself uses. Section 3.3 uses the priced objective for every controller that optimises a surrogate stage cost, which is every model-predictive controller except the full-model planner — that one maximises the simulator's own economic return and has no weights to reprice. The blocks generated by the supporting harness run under the original objective, and this is stated wherever it matters. Finally, closed-loop trajectories amplify platform-level floating-point differences over 5760 steps: the deterministic rule-based controller evaluates to -1.2061 EUR m^{-2} in the main harness and -1.2264 in the tuning harness under identical code and config_hash, so waves are compared within one environment and never cell-by-cell across environments.

3.1. One-step and multi-step criteria rank the feature libraries oppositely

The identification ladder evaluated 72 labelled configurations (three feature libraries × two polynomial degrees × four optimizers × three denoisers) at 20 seeds each, 1440 fits with no duplicates and a complete grid. Table 4 gives the degree-1, undenoised block and Figure 1 plots the two prediction criteria and the closed-loop outcome for all three libraries.

Adding physically motivated features monotonically inflates the regression condition number, from $\kappa = 8.21$ (raw) through 24.52 (physics without cross terms) to 53.43 (full physics), and monotonically degrades free-running prediction: the median 24-h rollout RMSE of the inside-air temperature rises from 3.31 to 7.11 to 30.84 °C, and the fraction of diverging rollouts from 0.000 to 0.011 to 0.130. The mean rollout error of the physics library, 1356.7 °C, is an artefact of divergent realisations (upper rollout quartile 2360.1); the median is the only defensible summary. κ depends on the library and the denoiser, not on degree or optimizer. The relation is equally clean at fixed optimizer, which matters because the closed-loop comparison below holds the optimizer fixed: under the ensemble estimator alone the median rollout errors are 2.67, 10.99 and 24.27 °C and the diverged fractions 0.0000, 0.0208 and 0.0767.

This is the whole of what conditioning is claimed to predict in this work. It orders the three libraries on open-loop multi-step stability, monotonically, and the ordering reproduces on a held-out year (below). It does *not* transfer to the closed loop. The same three libraries, compared one factor at a time — ensemble estimator, threshold 0.05, only the library changed — return +4.32, +0.28 and +2.75 EUR m^{-2} over four test seasons (Table 6), so the worst-conditioned library outscores the middle one by a factor of ten and the series in κ is non-monotone in closed loop. Nothing in this subsection is therefore an explanation of the economic result: the mechanism that does account for it is actuator-pathway survival, developed in Section 3.5.

The reversal against one-step accuracy is clean only for the two sparse estimators (STLSQ and ensemble, lower panel of Table 4): there the one-step ordering is strictly physics < physics_no_cross < raw (1.675 < 1.728 < 1.862 °C), the exact inverse of every multi-step ordering. Pooled over all four optimizer labels the one-step ordering is not strict — the raw library also beats physics_no_cross on one-step error (2.137 vs. 2.174). The supportable statement is therefore: *the library with the best one-step accuracy is the worst on every multi-step metric*, and the strict inversion holds under sparse estimation. Two properties of the grid qualify the counts: the constrained and sr3 labels are bit-identical in all 360 seed-cells, so 72 labels cover 54 distinct fits and the pooled upper panel of Table 4 double-weights that estimator; and embeddability coincides exactly with the degree-1 filter here (36 of 36 against 0 of 36), so it is not an independent screen. Of the 72 labels, 32 pass both pre-registered gates (embeddable and diverged fraction ≤ 0.05).

The frozen recipe used in the earlier version of this work (physics_no_cross/degree 1/ensemble/no denoising) passes both gates (diverged fraction 0.0208), but it ranks 8th of 72 on one-step error and 59th–60th on every multi-step metric. The comparator raw/degree 1/STLSQ/no denoising dominates it simultaneously on all three pre-registered axes: rollout RMSE 2.684 vs. 10.496 °C, diverged fraction 0.0000 vs. 0.0208, active terms 19.9 vs. 28.25. The frozen recipe is better on exactly one metric, one-step accuracy, which was never a gate. The full physics library at the same degree and estimator is worse still on the pre-registered axes — 63rd of 72 on median rollout error, 64th on divergence — and it is one of the 40 labels that *fail* the divergence gate (0.0767 > 0.05), so the open-loop procedure would have rejected it outright.

That last fact bounds what the selection rule may be claimed to do, and the bound belongs here rather than in the discussion. The rule identifies the winner: raw is first on multi-step stability and first in closed loop. It does not order the two losers. In closed loop the library the gate rejects scores +2.75 EUR m^{-2} and the library the gate passes scores +0.28 (Table 6) — the reverse of their open-loop ranking. The supportable claim is that multi-step stability is a better selection criterion than one-step accuracy and picks the best of the three libraries, not that it ranks libraries.

A held-out check (240 fits, 10 seeds × 3 libraries × 2 sparse optimizers × 4 fit→evaluate directions over 2018–2019) reproduces the library ordering in both regimes, with held-out/in-sample median ratios of ×1.04 (raw), ×1.27 (physics_no_cross) and ×0.99 (physics). It does not, however, isolate held-out status as the operative factor: pooled over fit year, median rollout error on 2018 versus 2019 data is 2.80/2.61 for raw, 16.94/2.70 for physics_no_cross and 27.19/15.85 for physics, and physics_no_cross narrowly beats raw in the 2018→2019 direction. The honest reading is that the raw library is stable irrespective of evaluation year whereas the physics libraries are strongly year-dependent.

Table 4. Identification ladder, degree-1 configurations without denoising. Upper panel pools the four optimizer labels ($n = 80$ fits per library); lower panel restricts to the two sparse estimators, STLSQ and ensemble ($n = 40$). Rollout RMSE is for the inside-air temperature at a 96-step (24-h) horizon. Source: regen/results/ladder_rerun/.

Library	κ	**Active**	**One-step**	**Rollout**	**Rollout**	**Diverged**
		terms	RMSE (°C)	median (°C)	IQR	fraction
			All four optimizer labels			
raw	8.21 ± 0.22	12.95	2.137	3.31	[2.67, 4.43]	0.000
physics_no_cross	24.52 ± 0.94	18.66	2.174	7.11	[4.24, 10.99]	0.011
physics	53.43 ± 2.40	26.74	**2.022**	**30.84**	[24.08, 2360.14]	0.130
raw	—	20.05	1.862	2.67	—	0.0000
physics_no_cross	—	28.18	1.728	10.58	—	0.0204
physics	—	36.98	**1.675**	**24.27**	—	0.0758

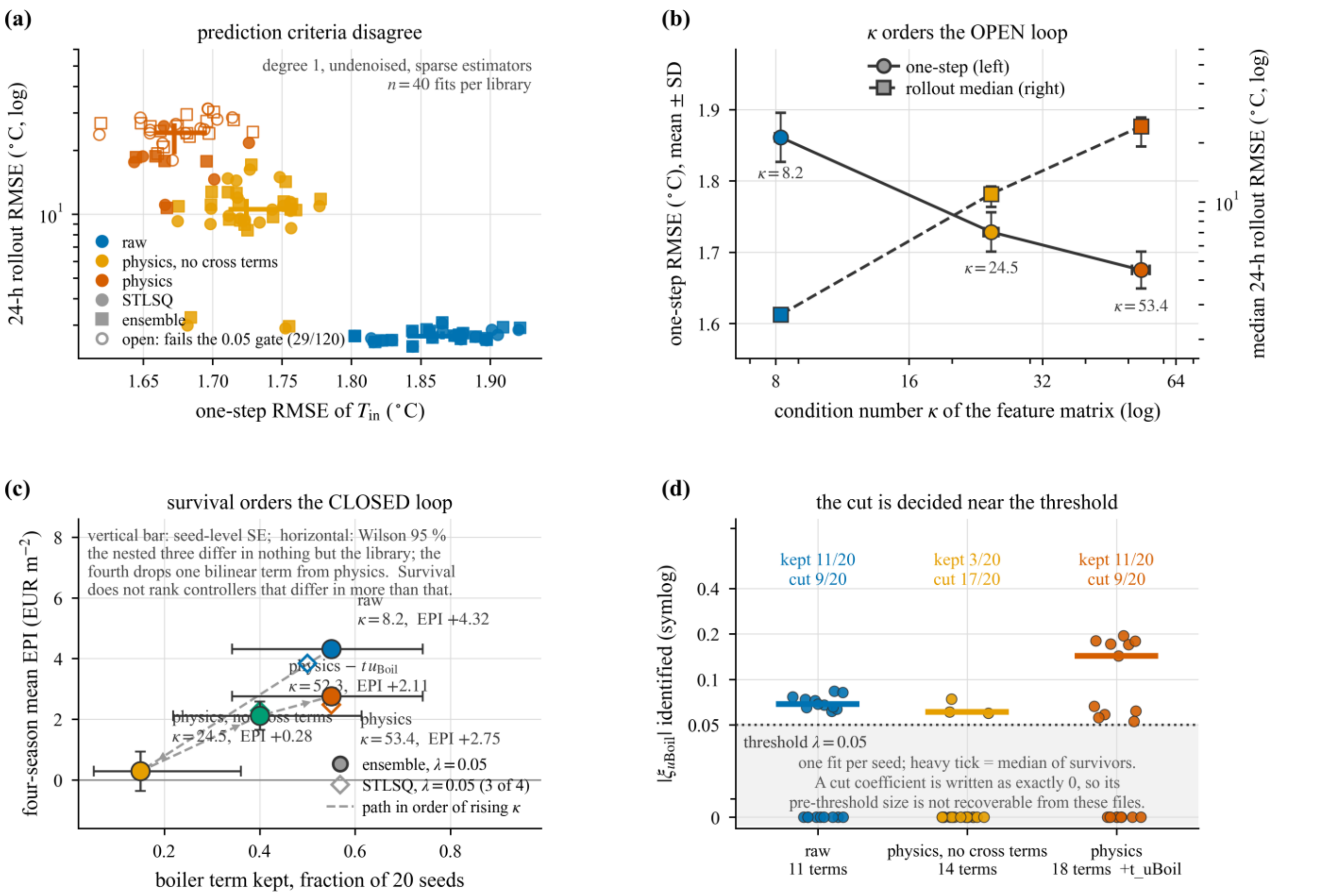


Figure 1. The selection criterion decides which library wins; conditioning does not decide by how much. (**a**) One-step RMSE of T_{in} against 24-h rollout RMSE, one marker per fit; open faces fail the 0.05 divergence gate, crosses give the per-library median and IQR. (**b**) The same block against the condition number of the regression: as physically motivated features are added, κ rises from 8.2 to 53.4, the one-step error falls and the median 24-h rollout error rises by an order of magnitude, so the two *open-loop* criteria order the three libraries in opposite directions. (**c**) The same three libraries in closed loop, changed one factor at a time (ensemble estimator, threshold 0.05), plus the 17-feature probe that deletes the single bilinear boiler term from the full physics set: four-season mean margin under the priced objective against the fraction of seeds in which the boiler-input coefficient survives thresholding. The closed-loop series is *not* monotone in κ — the worst-conditioned library outscores the middle one tenfold, and the probe, whose conditioning stays at the full library's level (κ = 52.3), does not collapse onto the middle library but lands between, in survival order — and the margin follows the survival axis instead (Section 3.5). **Scope:** the reversal in (**a**,**b**) holds in the degree-1, undenoised block under the two sparse estimators — 2 of 72 configurations per library, n = 40 fits per library. Pooled over all 72 labels the *raw* library has the best mean one-step RMSE (2.70 against 2.98 and 3.45) and there is no reversal. Data: regen/results/ladder_rerun/ for (**a**,**b**); priced_main/, priced_dagger/, phys_lib/ and notuboil/ (n = 80 per controller) for (**c**).

3.2. Tuning the rule-based reference

The rule-based controller had previously been described as tuned when in fact its setpoints were hard-coded. It was therefore tuned under the same 16-trial budget the reinforcement-learning agents are recorded as having received — 16 configurations drawn i.i.d. from a fixed generator, evaluated alongside the hard-coded configuration entered as trial 0, so 17 configurations were scored in all — and selected on the training seasons 2018–2019 only, the test seasons taking no part in selection. Table 5 reports the outcome. The predictive controllers received no hyper-parameter search of any kind (Section 2.3), so the budget is matched to the reinforcement-learning family alone.

Tuning is worth $+3.49$ EUR m^{-2} over four test seasons ($-1.23 \rightarrow +2.26$), and the gain is concentrated in the two adverse seasons, 2021 (+4.99) and 2023 (+5.79). The selected solution is agronomically interpretable — a wider diurnal swing (day setpoint 22.9 instead of 19.5 °C, night 14.4 instead of 16.5 °C) with a lower supplementary-lighting threshold — and it is bought with constraint pressure ($2813 \rightarrow 4339$ violation steps per season). Two facts qualify the exercise. Only 2 of the 16 random draws beat the hard-coded setpoints on the training score, placing the stock configuration at roughly the 88th percentile of the search space: it was not arbitrary, merely unsearched. And the heuristic is deterministic, giving $n = 4$ with no run-to-run dispersion; the spread across seasons is not an error bar.

Table 5. Tuning the rule-based reference under the reinforcement-learning agents' budget (16 random trials plus the hard-coded configuration as trial 0; selection on 2018–2019 only). EPI in EUR m^{-2}; one deterministic run per season. Source: regen/results/n2_tune/tune_rb_n2.csv.

Setpoint	**Range searched**	**Hard-coded**	**Selected**
Day temperature (°C)	17.0–23.0	19.5	22.92
Night temperature (°C)	14.0–20.0	16.5	14.38
Daytime CO_2 (ppm)	400–1200	800	828.3
Lamp radiation-sum limit	5.0–25.0	10	7.62
Test season	2020	2022	Mean Violations
Hard-coded	$+4.63$ -6.98	$+1.30$ -3.85	-1.23
Tuned	$+6.29$ -1.99	$+2.81$ $+1.94$	$\mathbf{+2.26}$
Gain	$+1.66$ $+4.99$	$+1.50$ $+5.79$	$\mathbf{+3.49}$ $+1526$

3.3. Four-season economic comparison

Fifteen controllers were compared on the test seasons 2020–2023 (Table 6; Figure 2). Every *surrogate-based* model-predictive controller ran under the priced objective at horizon 20. The reinforcement-learning controllers and the heuristic have no surrogate stage cost, and the full-model planner optimises the simulator's own economic return directly rather than a weighted surrogate cost, so none of the three was repriced and all are quoted from the canonical wave. Two of the fifteen — sindy_mpc_phys_ens and sindy_mpc_phys — carry the full physics library at the frozen recipe's threshold of 0.05 and complete the one-factor library series; they were run under the priced objective only, and they are the only two rows of the table with no counterpart under the original stage cost.

The raw-library SINDy-MPC leads with an ensemble optimizer ($+4.32$ EUR m^{-2}) and with STLSQ ($+3.83$), ahead of the two low-threshold physics_no_cross variants ($+3.09$, $+3.08$), the two full-physics variants ($+2.75$, $+2.48$), the tuned heuristic ($+2.26$), the on-policy re-identified variants ($+1.66$, $+1.37$), PPO [27] ($+0.45$), the frozen recipe ($+0.28$), the stock heuristic (-1.21), the full-model planner (-2.65, three seasons only), the neural-surrogate MPC (-3.82, 10 seeds) and SAC [28] (-4.15). The ranking of the nine SINDy controllers is therefore not ordered by feature library: the two full-physics controllers sit between the low-threshold physics_no_cross variants and the frozen physics_no_cross recipe, and sindy_mpc_phys_ens outranks sindy_mpc_conf by four places on a one-factor change of library alone — same degree, same denoiser, same optimizer, same threshold.

The advantage is not Pareto-dominant. On the plane (EPI ↑, violation steps ↓) the non-dominated set, recomputed over all fifteen controllers, still has exactly five members — sindy_mpc_raw_ens (+4.32 , 4130), sindy_mpc_lowthr (+3.09 , 3692), sindy_mpc_dense (+3.08, 3686), sindy_mpc_dense_dagger (+1.37, 3615) and ppo (+0.45, 1330). Neither new controller enters it: sindy_mpc_raw_ens dominates sindy_mpc_phys_ens (+2.75, 4193) and sindy_mpc_phys (+2.48, 4174) on *both* axes, higher margin at fewer violation steps, so the full physics library buys nothing that the raw library does not already provide. PPO attains roughly one third of the constraint pressure of the leader and is the only controller on the front that trades margin for safety at that rate. The tuned heuristic is *off* the front under the priced objective, dominated on both axes by sindy_mpc_dense. Front membership of sindy_mpc_dense_dagger is marginal: 7 of its 80 runs terminated early, and over completed runs only its violation mean rises to 3657 against 3683 for sindy_mpc_dense, narrowing the gap from 71 to 26 steps.

Paired tests against the tuned reference (Table 7, Holm correction over a family of 15, enlarged from 13 by the two new controllers) place four controllers significantly above it, all of them SINDy-MPC: both raw-library variants (Δ median +2.85 and +2.83, 75/80 and 68/80 wins) and both low-threshold physics_no_cross variants (+0.74, +0.69). Three controllers are not separable from the reference and must be described that way rather than as below it: sindy_mpc_phys_ens (Δ mean +0.49, median +0.25, 41/80, p_{Holm} = 0.176), sindy_mpc_phys (+0.21 , +0.29 , 44/80, 0.334) and sindy_mpc_conf_dagger (−0.60, −0.84, 31/80, 0.334). Every other controller in Table 7 is significantly below. The fifteenth member of the correction family, omitted from the table, is the stock heuristic re-evaluated inside the tuning harness: at $n = 4$ the signed-rank test cannot fall below $p =$ 0.125, so it is non-significant (Δ mean −3.49, 0/4, p_{Holm} = 0.334) even though its point estimate matches the significant rule_based row. Enlarging the family moved no controller across the 0.05 threshold, but every corrected level in the table rose, and the two levels quoted at 0.250 in the previous draft are now 0.334 for that reason alone. Because the tuned heuristic is deterministic with one run per season, its value is a constant reused across seeds: this is a one-sample signed-rank test against a constant, and the reference carries no sampling uncertainty. In a separate paired comparison between the two raw variants the test is null (Δ median −0.011, p = 0.60) — the library, not the optimizer, carries the effect. One caution on reading these levels: sindy_mpc_dense and sindy_mpc_lowthr are effectively one controller under the priced objective (per-run mean $|\Delta|$ = 0.026, four-season means 3.0822 vs. 3.0888), yet a paired test between them returns p = 0.012. Significance here is not magnitude, and the Holm family size is a reporting choice that shifts every corrected level.

The correction of the stage cost is itself a result. Matched on seed and season, the priced objective improves *both* axes for all eight MPC controllers for which both objectives were run, significantly in every case. The two full-physics controllers are not among those eight — they were introduced after the correction and exist only under the priced objective — so nothing here is claimed about them. The magnitude differs by an order of magnitude between the two libraries that were run twice: +2.74 and +2.68 EUR m^{-2} for sindy_mpc_dense and sindy_mpc_lowthr, with violations falling from ~6005 to ~3690, against +0.25 and +0.21 for the raw-library variants — a ratio of 10.9 to 12.8 depending on which pair is taken. Consequently the gap between the raw library and the best physics_no_cross controller narrows from +3.66 to +1.23 EUR m^{-2}: roughly two thirds of what previously looked like a library effect was a defective objective. The direction survives; the magnitude does not. Against the full physics library the priced gap is +1.56 (4.3173 against 2.7540), measured under one objective only and therefore not decomposable in the same way.

Table 6. Four-season economic comparison, EPI in EUR m^{-2} (mean ± SD over runs; n runs = seeds × seasons). MPC rows use the priced objective at horizon 20. The feature library of each SINDy-MPC row is given in the second column: raw is the 11 measured signals, p_nc adds three nonlinear transforms of the states (physics_no_cross), phys adds four bilinear state×input cross terms on top of those (physics). Sources: priced_main/, priced_dagger/, phys_lib/, final/main.csv, n2_tune/tune_rb_n2.csv.

Controller	Lib.	2020	2021	2022	2023	Four-season	n	Viol.
						mean ± SD		steps
sindy_mpc_raw_ens	raw	9.10 ± 0.36	−1.56 ± 0.58	4.62 ± 2.70	5.10 ± 0.39	**+4.32 ± 4.08**	80	4130
sindy_mpc_raw	raw	8.40 ± 2.24	−1.71 ± 0.70	4.04 ± 3.40	4.61 ± 1.59	+3.83 ± 4.23	80	4466
sindy_mpc_lowthr	p_nc	6.58 ± 1.25	0.27 ± 1.73	3.22 ± 1.35	2.29 ± 1.48	+3.09 ± 2.71	80	3692
sindy_mpc_dense	p_nc	6.58 ± 1.23	0.25 ± 1.71	3.21 ± 1.34	2.29 ± 1.45	+3.08 ± 2.70	80	3686
sindy_mpc_phys_ens	phys	6.35 ± 2.12	−1.91 ± 0.87	3.07 ± 2.34	3.51 ± 1.44	+2.75 ± 3.47	80	4193
sindy_mpc_phys	phys	5.82 ± 2.87	−2.02 ± 1.01	2.89 ± 2.76	3.22 ± 1.98	+2.48 ± 3.62	80	4174
rule_based_tuned	—	6.29	−1.99	2.81	1.94	+2.26	4	4339
sindy_mpc_conf_dagger	p_nc	4.56 ± 3.09	−1.73 ± 2.43	2.23 ± 3.74	1.57 ± 2.52	+1.66 ± 3.71	80	4138
sindy_mpc_dense_dagger	p_nc	4.74 ± 2.59	−1.59 ± 1.09	1.70 ± 1.78	0.63 ± 1.37	+1.37 ± 2.89	80	3615
ppo	—	3.36 ± 1.59	−2.10 ± 1.03	1.72 ± 1.19	−1.17 ± 3.40	+0.45 ± 2.97	80	**1330**

Controller	Lib.	2020	2021	2022	2023	Four-season	n	Viol.
sindy_mpc_conf	p_nc	3.88 ± 4.23	−3.16 ± 1.34	−0.31 ± 3.42	0.72 ± 3.46	+0.28 ± 4.11	80	4755
rule_based (stock)	—	4.63 ± 0.00	−6.97 ± 0.00	1.33 ± 0.00	−3.81 ± 0.00	−1.21 ± 4.52	80	2810
oracle_mpc	—	2.77 ± 0.08	−6.76 ± 0.09	—	−3.95 ± 0.06	−2.65 ± 4.03	60	3321
nn_mpc	phys*	−1.78 ± 1.95	−5.94 ± 1.99	−4.63 ± 1.82	−2.93 ± 1.73	−3.82 ± 2.42	40	2818
sac	—	−2.43 ± 1.33	−6.13 ± 1.47	−2.99 ± 1.61	−5.05 ± 1.20	−4.15 ± 2.05	80	1689

Notes. *nn_mpc is a dense neural surrogate fitted on the physics feature variant. It has no sparsity threshold and no identified coefficients, so the boiler-term survival used throughout Section 3.5 is undefined for it (xi_uboil is empty in all 40 of its rows, which must not be read as zero survival) and it is excluded from every survival comparison. oracle_mpc has no 2022 season: all 20 runs exhausted the common solver failure budget and were excluded, so its mean spans three seasons and is not directly comparable (see Section 3.6). nn_mpc has 10 seeds against 20 for all others. The two phys rows have 20 seeds, zero truncated seasons and zero solver failures. Both heuristics are deterministic: rule_based_tuned was run once per season (n = 4), and the stock heuristic was repeated over 20 seeds whose within-season SD is exactly zero. For both, the four-season dispersion entry is across-season spread and not run-to-run error, and the paired tests against them are one-sample tests against a constant. The stock heuristic is quoted from final/main.csv; the same controller evaluates 0.02 $\mathrm{EUR\,m^{-2}}$ lower (−1.23) and three violation steps higher (2813) in the tuning harness, and the tuning gain of Table 5 is computed within that harness, so the two values of this row are never mixed.

Table 7. Paired comparison against the tuned rule-based reference (priced objective). Δ = controller − tuned in $\mathrm{EUR\,m^{-2}}$; Wilcoxon signed-rank [30] with Holm correction [31] over a family of 15. The reference is deterministic, so this is a one-sample test against a constant. The family size is a reporting choice and shifts every corrected level: adding the two full-physics controllers raised all thirteen pre-existing levels without moving any of them across 0.05.

Controller	Δ mean	Δ median	Wins/n	p_{Holm}
sindy_mpc_raw_ens	+2.06	+2.85	75/80	8.3×10^{-11}
sindy_mpc_raw	+1.57	+2.83	68/80	6.5×10^{-7}
sindy_mpc_lowthr	+0.83	+0.74	51/80	7.0×10^{-4}
sindy_mpc_dense	+0.82	+0.69	51/80	7.0×10^{-4}

Controller	Δ mean	Δ median	Wins/n	p_{Holm}
sindy_mpc_phys_ens	+0.49	+0.25	41/80	0.176 (n.s.)
sindy_mpc_phys	+0.21	+0.29	44/80	0.334 (n.s.)
sindy_mpc_conf_dagger	−0.60	−0.84	31/80	0.334 (n.s.)
sindy_mpc_dense_dagger	−0.89	−0.79	23/80	1.2×10^{-4}
ppo	−1.81	−1.78	11/80	2.3×10^{-10}
sindy_mpc_conf	−1.98	−1.91	29/80	6.8×10^{-5}
rule_based (stock)	−3.47	−3.32	0/80	7.3×10^{-14}
oracle_mpc	−4.73	−4.79	0/60	1.8×10^{-10}
nn_mpc	−6.08	−5.52	0/40	2.4×10^{-11}
sac	−6.41	−6.68	0/80	1.1×10^{-13}

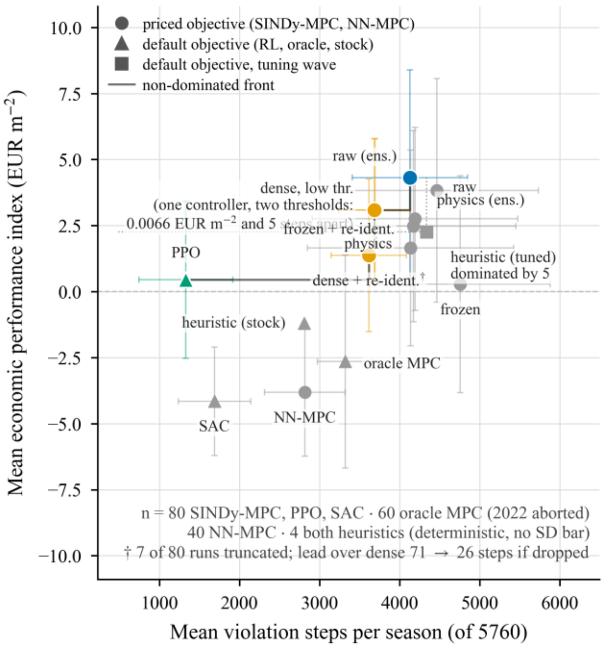


Figure 2. Margin against constraint pressure over four test seasons, priced objective, fifteen controllers. The non-dominated set has five members; the raw-library SINDy-MPC attains the highest margin and PPO the fewest violations, and the tuned heuristic lies inside the front. The two full-physics controllers are dominated on both axes by the raw-library leader — lower margin *and* more violation steps — so completing the conditioning series does not extend the front. Two of the five front points are one physics-informed controller under two thresholds, 0.0066 EUR m^{-2} and 5 violation steps apart, and a third owes part of its position to 7 truncated runs of 80. Replication is unequal: $n = 80$ for the SINDy and reinforcement-learning controllers, 60 for the full-model planner (2022 removed by the solver-abort rule), 40 for the neural-network MPC and 4 for either heuristic. Data: regen/results/priced_main/, priced_dagger/, phys_lib/, final/main.csv and n2_tune/tune_rb_n2.csv; the harnesses are kept apart deliberately.

3.4. Four unseen seasons

Four weather seasons (2014–2017) that took no part in identification, gate evaluation, selection or tuning were used as an external check (Table 8). These runs predate the objective correction and use the *original* stage cost; they are therefore not comparable in level with Table 6, only in ordering. The lead holds: the raw-library ensemble controller averages $+6.42\ \mathrm{EUR\,m^{-2}}$ ($\mathrm{SD} = 0.95$, $n = 32$) against $+2.38$ for the frozen recipe, $+1.07$ for the dense physics_no_cross variant and -0.33 for the hard-coded heuristic, and it wins in 32 of 32 seed×season pairs against each. With all differences of one sign at $n = 32$ the signed-rank test is at its exact floor, so these should be read as $p < 10^{-9}$ rather than as measured levels.

The ordering *below* the lead does not carry over. Compared like for like — the same four controllers on 2020–2023 under the same original objective — the dense variant scores $+0.34$, the frozen recipe -1.26 and the heuristic -1.21; on 2014–2017 the two physics_no_cross variants exchange places and the frozen recipe rises from last to second. Neither full-physics controller was run on these seasons, so this block speaks only to the two libraries it contains. Only the raw library's leadership and the heuristic's last place are invariant across the two sets of seasons, which is consistent with the year-dependence of the physics-informed libraries found in the held-out check (Section 3.1) and a further reason not to read any single ranking of the physics-informed variants as a property of the recipe. Three further caveats: only 8 seeds were run here against 20 in the main wave; the reference is the *hard-coded* heuristic, since the tuned setpoints were never evaluated on 2014–2017; and the margin is again bought with constraint pressure (4385 against 2624 violation steps).

Table 8. Four seasons never used at any stage, *original* objective, 8 seeds. EPI in $\mathrm{EUR\,m^{-2}}$. Source: regen/results/n5_years/main_n5.csv.

Controller	2014	2015	2016	2017	Mean	SD (n = 32)	Viol.
sindy_mpc_raw_ens	+7.08	+6.25	+5.98	+6.38	$\mathbf{+6.42}$	0.95	4385
sindy_mpc_conf	+3.06	+1.70	+2.18	+2.57	+2.38	4.03	5458
sindy_mpc_dense	+1.63	+0.55	+0.84	+1.27	+1.07	1.74	6226
rule_based (stock)	+0.99	−1.13	−2.22	+1.05	−0.33	1.42*	**2624**

* The heuristic is deterministic and its within-season SD is exactly 0.000 in all four seasons; the entry is across-season spread, not run-to-run error, and the paired test against it is one-sample, as in Table 6.

3.5. Mechanism

What determines the closed-loop margin is not the conditioning of the regression but whether the actuator pathway survives thresholding. Conditioning does predict one thing cleanly, and only that: open-loop multi-step stability, monotonically in the library (Section 3.1), as reported for ill-conditioned sparse regression [25] and as expected from the identification-for-control literature [16–17]. It does not predict the economics. The three libraries compared one factor at a time (Table 9) put the worst-conditioned basis in the *middle* of the economic ranking, ten times above the middle-conditioned one, so the closed-loop series in κ is non-monotone and no monotone conditioning argument can be made from it. The quantity that does track the ranking is the fraction of seeds in which the boiler-input coefficient $\xi_{u_{\text{boil}}}$ is retained by the sparsity threshold: the two libraries at 0.55 survival score $+4.32$ and $+2.75$ $\text{EUR}\,\text{m}^{-2}$, and the one library at 0.15 scores $+0.28$.

The comparison is one-factor by construction: degree, denoiser, optimizer and threshold (0.05) are held at the frozen recipe's values and only the feature library changes, over four seasons and 20 seeds with zero truncated runs. The same pattern appears in the STLSQ arm, where the raw library gives $+3.83$ (survival 0.50) and the full physics library $+2.48$ (survival 0.55); that arm has no four-season physics_no_cross controller at threshold 0.05, but the $\lambda = 0.05$ level of the sparsity sweep is exactly that configuration on season 2020, and it reproduces the anomaly — survival 0.10 and $+4.04$ $\text{EUR}\,\text{m}^{-2}$, against $+8.40$ (raw) and $+5.82$ (physics) on the same season. Reading across harnesses is what the environment caveat of the section preamble warns about, so this is offered as a consistency check on the shape and not as a fourth measurement.

Why the middle library is worse than either neighbour had one candidate explanation, and it has now been tested directly and rejected. The three libraries are nested: raw is the 11 measured signals; physics_no_cross adds saturation vapour pressure, vapour-pressure deficit and screen-attenuated radiation, which are nonlinear transforms of measured states and therefore near-collinear with them; physics adds four bilinear state × input terms on top, among them $T_{\text{in}} u_{\text{boil}}$. The candidate explanation held that the middle library carries enough collinearity for the threshold to destabilise the *direct* boiler term, while the full library — worse conditioned still — keeps the bilinear term as a detour able to carry the heating pathway once the direct one is cut. That account makes a sharp, falsifiable prediction: deleting exactly the bilinear feature from the 18-feature full library — sixteen of eighteen features, and essentially all of the collinearity, unchanged — should collapse the truncated library onto the middle one, with boiler survival near 0.15 and margin near $+0.28$ $\text{EUR}\,\text{m}^{-2}$, not leave it near $+2.75$ and 0.55.

The prediction fails (Table 9, fourth row pair). The 17-feature library (physics_no_tuboil), refit at both sparse estimators at threshold 0.05 and simulated over the same four seasons × 20 seeds with zero truncated runs, keeps the full library's conditioning ($\kappa = 52.3$ against 53.4, mean over 20 refits), its open-loop instability (median rollout 24.2 °C, divergence 0.084) and — decisively — its closed-loop level: $+2.11$ $\text{EUR}\,\text{m}^{-2}$ under the ensemble estimator and $+2.28$ under STLSQ. That is significantly above the middle library (mean paired difference $+1.83$, 62 of 80 pairs, $p_{\text{Holm}} = 4.4 \times 10^{-6}$ over the four contrasts of this block) and at most slightly below the full library (median -0.18, $p_{\text{Holm}} = 6.4 \times 10^{-3}$ under the ensemble estimator; the STLSQ arm is not separable, $p = 0.089$). Boiler-term survival moves only part of the way down, to 0.40 (8 of 20 replicates; exact McNemar test against the full library $p = 0.55$), and nowhere near the predicted 0.15. The small significant cost of the deletion under one estimator suggests the bilinear term shares the threshold's pressure with the direct term rather than substituting for it. With the detour ruled out, the margin ordering across the four libraries continues to track survival — $0.15 \to +0.28$, $0.40 \to +2.11$, $0.55 \to +2.75$ and $+4.32$ $\text{EUR}\,\text{m}^{-2}$ — while *which* feature or correlation pattern makes the threshold sever the direct term in physics_no_cross becomes an open question, stated as such in Section 4.

The controlled evidence for the mechanism is the single-coefficient ablation, and it is the only randomised element in this section. On the frozen recipe, paired over 20 seeds and one season, the boiler coefficient is re-inserted from the same seed's low-threshold fit with nothing else changed, lifting survival from 0.15 to 1.00. Under the priced objective the effect is significant ($p = 5.9 \times 10^{-4}$, $p_{\text{Holm}} = 1.2 \times 10^{-3}$ over the two knock tests, positive in 16 of 20 replicates) but typically small and heavily right-skewed: the median is $+0.21$ EUR m^{-2} and the mean $+1.92$, nine times the median and carried by a few seeds. All three statements belong together; the median is the defensible summary of the typical case and the mean is the defensible summary of the aggregate, and quoting either alone misrepresents the distribution. Under the original objective the same test gives a median of $+3.05$ at essentially the same significance, so the effect survives repricing while its typical size collapses by a factor of about 14 — most of the apparent value of the coefficient was an artefact of the mis-weighted stage cost, not of the term. Knock-out is not significant under either objective and its median is exactly zero, for the plain reason that the term is already absent in 17 of the 20 baseline seeds: there is usually nothing to remove. Cross-term interaction tests, run on the full physics library at threshold 10^{-6}, are uniformly non-significant and two of three point estimates flip sign between the objectives, so that block supports no direction. In particular it does not test the detour: at 10^{-6} the boiler term is retained in all 20 seeds, so the block removes coefficients from a model that never lost any, which is the opposite of the situation the detour is meant to explain.

Within one library the association is a dose-response. The sparsity sweep (Table 10 and Figure 3, physics_no_cross, season 2020 only, 20 seeds per level) shows the boiler term dying between $\lambda = 0.03$ (survival 0.95) and $\lambda = 0.06$ (survival 0) in both waves, whose survival profiles differ in one cell only ($\lambda = 0.04$: 0.35 priced against 0.30 original, one seed, and a consequence of the execution-environment drift noted above rather than of the objective). Levels at which at least one seed retains the term average $+5.77$ EUR m^{-2} against $+3.47$ for levels at which none does, under the priced objective ($+3.46$ vs. $+2.77$ under the original). The separation is not monotone under either objective — the worst retaining level (4.04, priced) is below the best non-retaining one (4.61) — so the statement "EPI is monotone in boiler-term survival" is false as written, here as it is across libraries. The priced weights raise EPI at every one of the 13 levels ($+0.34$ to $+2.95$) and cut violations at every level.

Table 11 decomposes EPI over the four test seasons by library and by whether the boiler term survived, now over all three libraries. Inside physics_no_cross the gap is $+2.34$ EUR m^{-2} and inside raw $+1.13$; inside physics it is -0.22 and not significant. The raw library with the term *dropped* ($+3.48$) still beats every cell of both physics libraries, including physics_no_cross with the term *kept* ($+2.36$), so survival does not exhaust the library effect either. This decomposition is observational, not a randomised ablation — survival is a property of the fit, constant across the four seasons of a seed, so the effective sample is 20 seeds per controller and not 80 runs, and the cells differ in composition as well as in the factor of interest: under the priced objective the "kept" cell of physics_no_cross is filled by the three recipes that never lose the term (80 runs each), plus most of one re-identified variant (68 of 80), plus the 12 runs in which the frozen recipe happens to keep it; its "dropped" cell is 68/80 the frozen recipe. Between seeds the boiler association is significant only inside physics_no_cross ($p = 0.0015$ priced, $p = 0.0006$ original) and not inside raw ($p = 0.136$, $p = 0.316$) or physics ($p = 0.178$). The causal estimate remains the knock-in median of $+0.21$, not the -0.2 to $+2.3$ spread the three observational splits suggest.

Four limits qualify the whole of this subsection. The randomised ablation is single-season (2020) and single-recipe, so the causal estimate is not measured under the seasonal variation that the rest of the results report. The knock and cross blocks are anchored on different baselines (survival 0.15 against 1.00) and on different libraries, so their deltas are not one ablation family. The library comparisons — including the term deletion, which is a structural edit and not an assigned factor — are randomised in the seed but not in the library: libraries are not assigned, they are chosen, so these are controlled comparisons rather than experiments on the mechanism. And survival is measured on one coefficient of one actuator; whether the same account holds for the lamp and CO_2 pathways was not tested.

Table 9. One-factor library comparison. Degree, denoiser and sparsity threshold (0.05) are held at the frozen recipe's values; only the feature library changes. Open-loop columns are degree-1 undenoised ladder fits at the matching optimizer (n = 20); closed-loop columns are four test seasons × 20 seeds under the priced objective (n = 80, no truncated runs). ξ survival is the fraction of runs retaining the boiler-input coefficient. The p_ntb rows are the 17-feature term-deletion library that tests the bilinear-detour reading (Section text): deleting the one cross term that touches the boiler leaves κ, rollout and divergence at the full library's level, and the predicted collapse onto the p_nc row does not occur. Sources: regen/results/ladder_rerun/, priced_main/, priced_dagger/, phys_lib/, notuboil/.

Library	Controller	κ	Rollout	Diverged	ξ	EPI
			median (°C)	fraction	survival	(four-season)
			Ensemble estimator			
raw	sindy_mpc_raw_ens	8.21	**2.67**	**0.0000**	0.55	**+4.32 ± 4.08**
p_nc	sindy_mpc_conf	24.52	10.99	0.0208	0.15	+0.28 ± 4.11
phys	sindy_mpc_phys_ens	53.43	24.27	0.0767	0.55	+2.75 ± 3.47
p_ntb	sindy_mpc_notuboil_ens	52.28	24.17	0.0842	0.40	+2.11 ± 3.51
raw	sindy_mpc_raw	8.21	2.67	0.0000	0.50	+3.83 ± 4.23
p_nc	— (not run)	24.52	10.45	0.0200	—	—
phys	sindy_mpc_phys	53.43	24.49	0.0750	0.55	+2.48 ± 3.62
p_ntb	sindy_mpc_notuboil	52.28	23.58	0.0783	0.40	+2.28 ± 3.55

The first three columns are monotone across the three nested libraries and the last two are not: the worst-conditioned library outscores the middle one by a factor of ten in closed loop, and the economic ranking follows ξ survival rather than κ; the p_ntb row sits between the two physics rows by construction and is excluded from that statement. No four-season controller exists for physics_no_cross under STLSQ at threshold 0.05; the $\lambda = 0.05$ level of Table 10 is that configuration on season 2020 and gives survival 0.10 at +4.04 EUR m^{-2}, against +8.40 and +5.82 for the raw and phys STLSQ rows above on the same season. That is a cross-harness comparison and is quoted as a consistency check only.

Table 10. Sparsity sweep within the physics_no_cross library, season 2020, 20 seeds per level. ξ survival is the fraction of seeds retaining the boiler-input coefficient; it and the active-term count are quoted from the priced wave. Identification precedes the closed loop and does not depend on the objective, but the two waves were executed in different environments, so their fits are not bit-identical: 28 of the 260 (λ, seed) cells differ by one or two active terms, level means by at most 0.15, and one cell flips ξ survival — whence the original-objective survival at $\lambda = 0.04$ is 0.30 rather than 0.35. EPI in $\text{EUR}\,\text{m}^{-2}$. Sources: regen/results/priced_mech/ and regen/results/final/mechanism*.csv.

λ	Active terms	ξ survival	EPI (priced)	EPI (original)	Δ
10^{-6}	53.95	1.00	6.58 ± 1.25	3.68 ± 1.85	$+2.90$
10^{-3}	49.60	1.00	6.58 ± 1.23	3.63 ± 1.93	$+2.95$
0.01	42.65	1.00	6.11 ± 2.52	3.21 ± 3.25	$+2.90$
0.02	38.80	1.00	6.38 ± 2.01	4.39 ± 1.47	$+1.99$
0.03	36.10	0.95	6.41 ± 1.82	4.98 ± 1.76	$+1.43$
0.04	32.15	0.35	4.30 ± 3.51	2.76 ± 3.84	$+1.54$
0.05	28.10	0.10	4.04 ± 4.14	1.59 ± 5.59	$+2.45$
0.06	25.45	0.00	3.59 ± 2.96	2.58 ± 4.63	$+1.01$
0.07	23.05	0.00	4.01 ± 2.37	3.67 ± 2.62	$+0.34$
0.08	21.30	0.00	3.04 ± 1.09	2.49 ± 1.23	$+0.55$
0.10	19.50	0.00	3.11 ± 0.36	2.50 ± 0.34	$+0.61$
0.15	14.40	0.00	4.61 ± 0.93	4.05 ± 0.99	$+0.56$
0.20	12.15	0.00	2.43 ± 2.49	1.34 ± 3.05	$+1.09$

Table 11. Mean EPI ($\text{EUR}\,\text{m}^{-2}$) by feature library and boiler-term survival, pooled over the nine SINDy-MPC controllers and four test seasons; run counts in parentheses. The physics library was run under the priced objective only. Sources: final/main.csv, n7/main_n7.csv (original objective), the priced pool and phys_lib/.

	Original objective		**Priced objective**		**Boiler**
Library	dropped	kept	dropped	kept	effect (priced)
raw	$+3.20$ (76)	$+4.43$ (84)	$+3.48$ (76)	$\mathbf{+4.61}$ (84)	$+1.13$ (n.s.)
physics_no_cross	-1.42 (88)	$+0.15$ (312)	$+0.02$ (80)	$+2.36$ (320)	$\mathbf{+2.34}$
physics	—	—	$+2.74$ (72)	$+2.52$ (88)	-0.22 (n.s.)
Library effect, raw − physics_no_cross	$+4.62$	$+4.28$	$+3.46$	$+2.25$	

	Original objective		Priced objective		Boiler
Library effect, raw − physics	—	—	+0.75	+2.09	

Survival is a property of the fit and is constant across the four seasons of a seed, so the effective sample is 20 seeds per controller and not the printed run count. Between seeds the association is significant only inside physics_no_cross ($p = 0.0015$ priced, $p = 0.0006$ original) and not inside raw ($p = 0.136$, $p = 0.316$) or physics ($p = 0.178$). The last of those three nulls is the observation that motivates the cross-term account in the text; it is not a test of it.

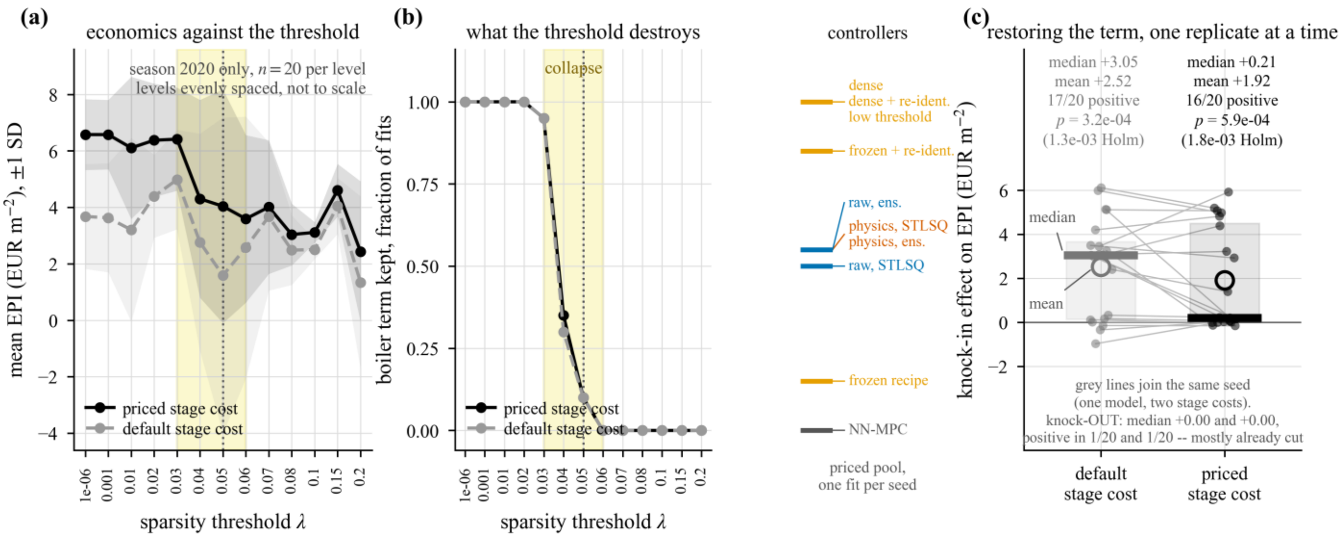


Figure 3. Sparsity sweep, boiler-term survival and the controlled single-coefficient ablation; physics_no_cross library, season 2020, 20 seeds per level. (**a**) Mean margin against the sparsity threshold λ under both objectives, ± 1 SD shaded; the priced objective lifts the margin at every level. (**b**) Survival fraction of the boiler coefficient over the same grid: it disappears between $\lambda = 0.03$ and $\lambda = 0.06$ under both objectives. Ticks give per-controller survival in the main comparison, from 0.15 for the frozen recipe to 1.00 for the low-threshold variants, with 0.50 and 0.55 for the two raw-library controllers and 0.55 for both full-physics controllers; the neural-surrogate MPC has no thresholded coefficients and therefore no tick. (**c**) Knocking the coefficient back in, per replicate and paired on seed, under each objective; the heavy tick is the median and the open symbol the mean. Under the priced objective the effect is significant but typically small and heavily right-skewed — median $+0.21$ EUR m^{-2} against a mean of $+1.92$, positive in 16 of 20 replicates. Data: regen/results/priced_mech/ and regen/results/final/mechanism*.csv.

3.6. Alternative explanations

Model accuracy is not the binding constraint. A planner using the simulator's own dynamics ranks in the lower third (-2.65 EUR m^{-2}), and its shortfall cannot be attributed to the objective correction of Section 3.3: it never optimised the weighted surrogate cost in the first place, but maximised the simulator's own economic return over its horizon. What it does have is a weak optimiser — a warm-started cross-entropy search over a few dozen candidate sequences — so the result is about optimisation quality, not model quality. Its 2022 season is absent from Table 6 because all 20 runs hit the common solver-failure budget of 100, at about 95 % of the season (season fraction 0.936–0.987, mean 0.950). This is a budget artefact, not infeasibility: with the budget raised to 10^5 the season completes on both tested seeds, requiring 140 and 103 failures and yielding -1.12 and -0.56

(mean -0.84). Imputing that mean gives a four-season figure of about -2.20 instead of -2.65; both should be quoted, and the correct phrasing is that the full model does not fit the common budget, not that it cannot be solved. Its rank is unaffected either way.

Prediction error along the operating trajectory is small. Replaying both physics_no_cross surrogates along the full-model planner's own closed-loop trajectory (5760 steps, 5 seeds, season 2020) gives one-step temperature RMSE of 1.89 ± 0.09 and 1.82 ± 0.04 °C ($R^2 = 0.82$ and 0.83) and 24-h rollout RMSE of 2.62 and 2.49 °C, with zero diverging rollouts out of 100 at every horizon. The same frozen recipe measures about 10.5 °C on the excitation data of the ladder — roughly four times larger. The surrogate is far more accurate along a smooth closed-loop trajectory than on PRBS-excited identification data, and the ladder number must not be read as its error in operation.

The horizon was never tuned, and correcting for it does not change the ordering. At horizon 8 every MPC improves over horizon 20 on matched seeds and seasons (sindy_mpc_raw_ens $+4.58$ vs. $+4.29$, $\Delta = +0.29$, 33/40; sindy_mpc_lowthr $+2.78$ vs. $+0.67$, $\Delta = +2.11$, 40/40; sindy_mpc_dense $+2.65$ vs. $+0.53$, $\Delta = +2.12$, 40/40; sindy_mpc_conf $+0.90$ vs. -0.40, $\Delta = +1.30$, 37/40), with no rank change. The size of the gain is inversely related to multi-step prediction quality — $+0.29$ for the raw library against $+1.3$ to $+2.1$ for the physics_no_cross variants — which is exactly the compounding-error mechanism. The two full-physics controllers were not in this block, so this argument is not extended to them. A single-controller horizon sweep confirms the direction (6.11, 5.13, 3.63, 0.91 EUR m^{-2} at $h = 8{,}12{,}20{,}30$, with dispersion growing from SD $= 0.88$ to 4.03). The asymmetry runs against the MPCs: the learned controllers received 16 tuning trials and the horizon was never tuned for any MPC.

The bootstrap draw of the ensemble optimizer is a real but secondary axis. Resampling the ensemble draw (6 seeds $\times$ 6 draws $\times$ 4 seasons, original objective) leaves the raw-library controller leading in 32 of 36 seed×draw cells (0.89). Its spread across draws within a seed averages 0.26 EUR m^{-2} against 1.11 for the frozen recipe and 2.04 for the re-identified variant, but that mean is not representative: five of the six seeds lie at or below 0.25 and one seed alone contributes 1.18. The controller has one bad basin rather than uniform insensitivity to the draw. This wave disagrees with the canonical 20-seed, 10-draw wave on the frozen recipe's boiler survival (0.00 vs. 0.12) and on the leadership ordering, so the two must be cited separately and never merged.

3.7. Adaptation and safety

Three adaptation modes were compared on the three harder seasons (2021–2023) with 20 seeds $\times$ 10 ensemble draws per condition, 600 runs each, under the original objective. On-policy re-identification — the surrogate is rolled out in closed loop, the visited states are appended to the training set and the model is re-estimated, with no expert queried, so this is not DAgger [38] — gives -0.39 EUR m^{-2} against -1.58 for the static model (paired $\Delta = +1.19$, median $+0.41$, 380/600, $p = 5.0 \times 10^{-16}$) and reduces violations from 5424 to 4654. Recursive coefficient adaptation by extended Kalman filter is harmful: -2.95 EUR m^{-2} ($\Delta = -1.36$ against static, $p = 7.4 \times 10^{-14}$) with 234 of 600 seasons terminating early. All three are net-negative in absolute terms because 2020, the favourable season, is absent from this block.

The Mahalanobis-distance safety guard [39–40] does not pay. Guarded runs average -2.59 EUR m^{-2} against -1.91 unguarded; the paired mean difference is -0.69 but the paired median is $+0.19$ with the guard better in 335 of 600 pairs ($p = 0.0047$), because the guard shortens a heavy left tail while shaving the centre; both statistics must be reported. Violations are *worse* under the guard ($+430$ steps, worse in 529 of 600, $p = 5 \times 10^{-85}$), and 179 of 600 guarded seasons terminate early; restricted to completed seasons the margin difference is -0.33 and not significant ($p = 0.081$). The underlying detector is weak: the area under the ROC curve is 0.663 ± 0.042 for the distance statistic and 0.627 ± 0.058 for ensemble spread [41], barely above chance. The guard's activation count was never recorded, so the natural explanatory variable is unmeasured. This is a negative result and is reported as one.

The residual-based fault supervisor, in contrast, works (Table 12). It improves the margin under all six injected faults (paired over 20 seeds, all $p \leq 1.7 \times 10^{-3}$, and unchanged in that bound under Holm correction over the six) and collapses the dispersion from $\mathrm{SD} = 1.16$–5.48 to 0.29–0.39: it buys predictability more than margin. Violations fall sharply (9131 to 5292 under a dead vent; 5198 to 2660 under a stuck boiler). Two faults remain net-destructive even when supervised. One result deserves plain statement rather than burial: with the lamps dead, the supervised fallback (+6.33) outperforms the nominal fault-free controller (+3.63), which implies that the nominal MPC over-uses supplementary lighting.

Table 12. Fault injection with and without the residual-based supervisor, season 2020, 20 seeds. EPI in EUR m^{-2}, mean (SD); paired Wilcoxon over seeds. The fault-free reference is +3.63 (SD 1.93). Source: regen/results/final/tables/faults.csv.

Fault	No supervisor	Supervised	Δ	Wins	vs. fault-free
uVent_dead	−18.19 (5.48)	+0.10 (0.30)	+18.29	20/20	−3.53
uBoil_stuck	−8.41 (5.35)	−3.88 (0.29)	+4.54	16/20	−7.50
rh_offset	+1.27 (3.72)	+4.45 (0.36)	+3.19	20/20	+0.83
uLamp_dead	+3.50 (1.64)	+6.33 (0.29)	+2.82	20/20	+2.70
t_in_stuck	+3.06 (1.18)	+4.42 (0.29)	+1.36	17/20	+0.79
t_in_offset	+3.67 (1.16)	+4.47 (0.39)	+0.80	17/20	+0.84

3.8. Sensitivity to design parameters and to prices

Perturbing the identified coefficients degrades the closed loop non-uniformly (Table 13, Figure 4). Mean EPI falls from $+4.11$ at a $2\,\%$ perturbation to -8.13 at $20\,\%$, a span of 12.24 EUR m^{-2}, while the median falls only from $+4.47$ to -0.31, a span of 5.72: degradation takes the form of a growing minority of catastrophic runs, not a uniform decline. Only the two largest levels are significant against the smallest on per-seed means ($p = 0.0059$ at $15\,\%$, $p = 0.0020$ at $20\,\%$; $p > 0.6$ below that), and the $2\,\% \rightarrow 5\,\%$ step is already inverted. The canonical coarser grid, which also covers $30\,\%$, gives $+1.67$, -13.41, -4.77 at $10/20/30\,\%$ — a span of 15.08 and non-monotone; the finer rerun stops at $20\,\%$, so that non-monotonicity is unreproduced rather than refuted. Either way, the sensitivity is 4.5–5.6 times larger than the value of 2.7 EUR m^{-2} claimed in the superseded manuscript. Varying the sparsity threshold of the same controller spans 5.00 EUR m^{-2} on this 10-seed rerun but only 1.87 on the canonical 20-seed grid; the two disagree by about 1 EUR m^{-2} even on matched seeds at the two lowest thresholds, so the span of this factor is not well determined and only the direction should be relied on. Violations rise by half between the extreme levels (6167 at threshold 0.01 against 9570 at 0.20) but *not* monotonically: the 0.05 level, at which the boiler term is lost in 9 of 10 seeds, has the fewest violations of the four (5155), and the canonical grid reproduces that dip (6263, 5423, 8440, 9219). Varying the horizon spans 4.76–5.21.

Re-scoring the recorded physical quantities over a 3×3 price grid (fruit price $\times$ energy scale, each spanning a factor of four) yields per-controller spans of 31.5 to 50.2 EUR m^{-2}, median 38.3. Prices therefore dominate design parameters numerically, but the axes are not commensurable — a factor of four in prices against $\pm 20\,\%$ on coefficients — and no claim that "prices matter more than design" is supportable from this comparison. What is supportable is that the ranking is not price-invariant either: two distinct controllers win across the nine price points (the heuristic at six, a re-identified SINDy variant at three). This last statement carries an important restriction: the price grid was re-scored only for the ten controllers of the canonical wave, and *neither raw-library controller is in it*. The price sensitivity of the leading controller of Table 6 has therefore not been measured, and the nine-point winner counts must not be read as a re-ranking of the main comparison.

One provenance defect must be disclosed here. The supporting harness that generates the design and sensitivity blocks hard-codes the original objective in every rollout call; the objective flag is consumed only by the main and mechanism blocks, so the reruns stored under a priced label in this block are *not* priced. The evidence is direct rather than inferential. The mechanism sparsity sweep and the design threshold factor fit the same model at the same levels, and under the original objective the two agree exactly on all 80 overlapping (level, seed) cells (max $|\Delta| = 0$). The rerun labelled priced, by contrast, disagrees with the genuinely priced mechanism sweep on *all* 40 overlapping cells (mean $\Delta = +1.23$ EUR m^{-2}), while remaining within execution-environment drift of the original-objective values (39 of 40 horizon cells and 36 of 40 threshold cells within 0.2 EUR m^{-2}; the remainder are chaotic amplifications). The coefficient and threshold sensitivities have consequently never been measured under the priced objective.

Table 13. Coefficient-perturbation sensitivity of a single SINDy-MPC controller, season 2020, 10 seeds $\times$ 4 repetitions, original objective. EPI in EUR m^{-2}. Source: regen/results/priced_design/design_pricedDesign*.csv (deduplicated).

Perturbation	Mean	SD	Median	Min	Early terminations
0.02	+4.11	1.29	+4.47	+1.25	0/40
0.05	+4.03	2.06	+4.35	−1.19	0/40
0.10	+2.96	7.57	+5.40	−34.18	0/40
0.15	−3.86	14.27	+1.12	−42.87	4/40
0.20	−8.13	18.39	−0.31	−42.16	12/40
Span	12.24	—	5.72	—	—

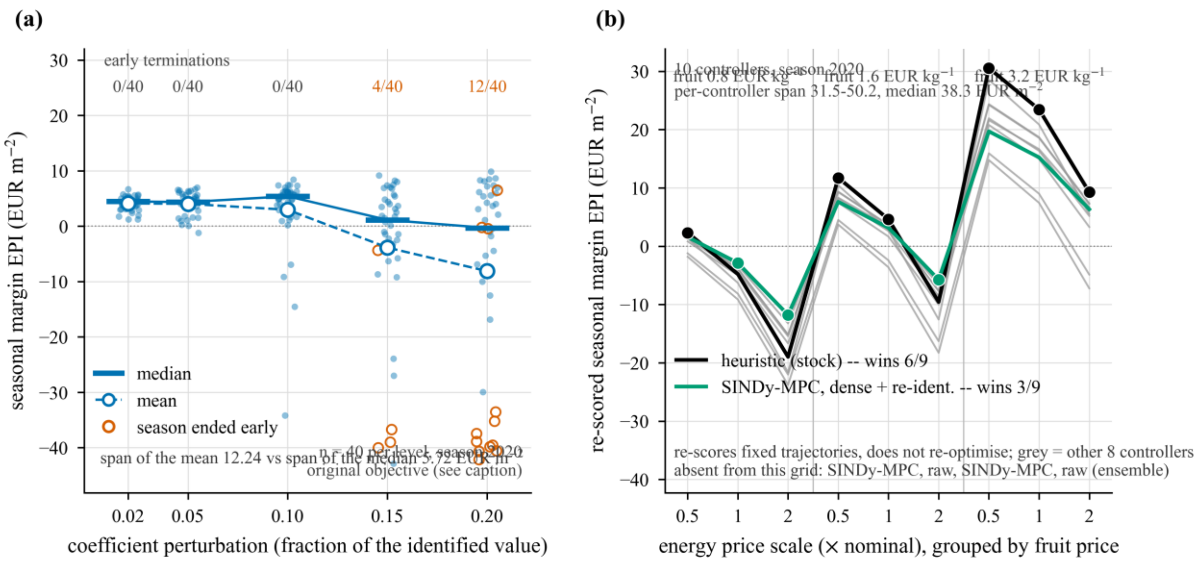


Figure 4. Sensitivity of the seasonal margin. (**a**) Run-level margin under multiplicative perturbation of the identified coefficients, season 2020, 40 runs per level; the heavy tick is the median, the open symbol the mean, and early terminations are counted per level. The mean falls by 12.2 EUR m^{-2} across the grid while the median falls by 5.7: the degradation is a growing minority of catastrophic runs rather than a uniform decline. **This panel is a default-objective result** — the supporting harness hard-codes the original stage cost, so the directory name priced_design/ is a misnomer. (**b**) Margin over a nine-cell grid of fruit price × energy scale. The grid *re-scores* fixed trajectories rather than re-optimising them, and covers only the ten canonical-wave controllers: **neither raw-library controller appears in it**, so it does not test the ranking reported here. Data: regen/results/priced_design/ (**a**) and regen/results/final/main.csv (**b**).

4. Discussion

4.1. Why physically motivated features can hurt, and why the damage is not monotone in how many are added

The physically motivated regressors used here—saturation vapour pressure $e_s(T_{\mathrm{in}})$, vapour-pressure deficit D and screen-attenuated radiation $I(1-u_{\mathrm{ThScr}})$—are algebraic functions of variables the raw library already contains: no new excitation, only smooth near-collinear copies of existing columns. Adding them, and then the four bilinear cross terms, raises the condition number of the regression monotonically, and open-loop multi-step stability degrades monotonically along the same direction. Over the ladder's degree-one undenoised block (Section 3.1), $\kappa = 8.2 \rightarrow 24.5 \rightarrow 53.4$, median 24 h rollout error $3.31 \rightarrow 7.11 \rightarrow 30.84$ °C and divergence $0.000 \rightarrow 0.011 \rightarrow 0.130$. One-step error does not order the libraries along that direction at all: it is *best* for the worst-conditioned library (2.02 °C) while the middle library is marginally worse than raw (2.17 against 2.14), so what the ladder establishes is a contrast between the extreme points, not a monotone reversal. Table 14 shows the same contrast on the held-out identification block, where the three libraries score $\kappa = 8.2$, 25.4 and 56.2; that block is a different sample of fits (10 replicates per fit year rather than 20 on both), so the two sets of figures must not be mixed within one comparison.

That much is a statement about open-loop prediction, and it holds. An earlier version of this argument continued the chain into the closed loop: collinearity leaves the least-squares solution weakly determined along the ill-conditioned directions, coefficients become unstable across identification replicates, a fixed sparsity threshold arbitrates inconsistently between near-equivalent representations, and the resulting rollout error compounds into economic loss. *The second half of that chain does not survive measurement.* When the extreme point of the conditioning series was finally carried into closed loop—ensemble optimiser, threshold 0.05, priced objective, four seasons × 20 replicates, zero truncated runs, the feature library the only thing that changes—the three libraries scored +4.32 (raw), +0.28 (physics without cross terms) and +2.75 EUR m^{-2} (full physics). The series is **non-monotone in** κ: the worst-conditioned library beats the middle one by +2.47 EUR m^{-2} on average (median +2.39, 63 of 80 paired runs, $p = 1.3 \times 10^{-8}$), and it does so in each of the four seasons taken separately. Conditioning therefore predicts open-loop multi-step stability cleanly and closed-loop economics not at all. The agreement observed while only two libraries had closed-loop controllers was a two-point coincidence, and the third point destroys it.

What the closed-loop ordering does track is whether the boiler term survives thresholding. The direct coefficient $\xi_{u\mathrm{Boil}}$ is retained in 11 of 20 identification replicates with the raw library, 11 of 20 with the full physics library, and 3 of 20 with the physics library without cross terms; the two libraries at 11 of 20 score +4.32 and +2.75, and the one at 3 of 20 scores +0.28. Restated as a mechanism: what matters is not how well conditioned the regression is, but whether the pathway through which an actuator reaches the controlled state survives sparsification. A surrogate whose boiler column has been thresholded away cannot be steered with the boiler, however accurate it is one step ahead and however well conditioned the fit that produced it.

The controlled evidence for that reading is the single-coefficient ablation, which this result promotes from a supporting measurement to the central one: survival explains the closed-loop outcome and κ does not. Reinstating the deleted coefficient in the frozen recipe is the only randomised manipulation in the mechanism block—library, scalers, every other term and the MPC construction held fixed, one number changed. Under the priced objective the effect is significant and typically small: positive in 16 of 20 replicates, $p = 5.9 \times 10^{-4}$ (1.2×10^{-3} after Holm correction over the two knock tests; 1.8×10^{-3} if the family is widened to the four mechanism contrasts, as in Figure 3), median +0.21 EUR m^{-2} with interquartile range [+0.02, +4.49], mean +1.92 (Figure 3c). The mean is nine times the median because a minority of replicates carry it, so the median is the honest summary; the +3.05 median obtained under the defective objective is superseded and must not be quoted as the size of this effect.

Survival gates the outcome; it does not order it, and two measurements say so. Among the two libraries that keep the term, the raw one still wins by +1.56 EUR m^{-2} (median +0.97, 67 of 80, $p = 2.5 \times 10^{-9}$)—and it is also the one with the better rollout, so the open-loop criterion of Section 4.2 separates the two survivors where survival does not. And in the decomposition over (library) × (term retained), the raw library *without* the term (+3.48 EUR m^{-2}) still beats the physics library *with* it (+2.36). That split is observational—survival is a property of each fit, not a randomised factor, and it is confounded with recipe—so the conservative reading is that survival marks which fits collapse rather than measuring the whole library effect.

Why the middle library should be worse than *both* of its neighbours had one candidate explanation, and the experiment that decides it has now been run. The explanation was that the library without cross terms carries enough collinearity for the threshold to destabilise the direct boiler term, but—unlike the full library—holds no bilinear $T_{\mathrm{in}}u_{\mathrm{Boil}}$ term able to carry the heating pathway once the direct term is cut: the worse-conditioned library has a detour, the middle one has neither stability nor a detour. Two observational figures were consistent with it, and neither is significant: inside the full physics library, replicates that lose the direct term score no worse than those that keep it (+3.10 against +2.48 EUR m^{-2}, 9 and 11 replicates, $p = 0.25$), whereas inside the library without cross terms losing it costs +2.26 (+2.20 against −0.06, 3 and 17 replicates, $p = 0.18$).

The term-deletion test rejects the detour. Deleting exactly the bilinear $T_{\mathrm{in}}u_{\mathrm{Boil}}$ feature—the only one of the four cross terms that touches the boiler—from the full physics candidate set gives a 17-feature library that keeps sixteen of eighteen features and essentially all of the conditioning ($\kappa = 52.3$ against 53.4, mean over the 20 refits). Refit at the same optimiser and threshold and simulated over four seasons $\times$ 20 replicates in closed loop, it does not fall back towards the $+0.28\ \mathrm{EUR\,m^{-2}}$ and 0.15 survival of the library without cross terms, which is what a bilinear detour predicts. It scores $+2.11\ \mathrm{EUR\,m^{-2}}$ under the ensemble estimator and $+2.28$ under STLSQ, with survival 0.40 (8 of 20 replicates, Wilson interval $[0.22, 0.61]$; exact McNemar test against the full library $p = 0.55$): significantly above the middle library (mean paired difference $+1.83$, 62 of 80 pairs, $p_{\mathrm{Holm}} = 4.4 \times 10^{-6}$, Holm correction over the four contrasts of this block) and at most slightly below the full library (median -0.18, $p_{\mathrm{Holm}} = 6.4 \times 10^{-3}$ under the ensemble estimator; not separable under STLSQ, $p = 0.089$). The small significant cost of the deletion in one estimator arm suggests the bilinear term shares the threshold's pressure with the direct term rather than replacing it. Which feature, or which correlation pattern, makes the threshold sever the direct term in the library without cross terms is therefore open; what the mechanism claims is untouched, and is in fact sharpened by the fourth point: across four libraries the margin continues to track survival—$0.15 \to +0.28$, $0.40 \to +2.11$, $0.55 \to +2.75$ and $+4.32\ \mathrm{EUR\,m^{-2}}$—while neither conditioning, one-step accuracy nor library size orders them. The cross-term block already present in the mechanism experiment could not have substituted for this wave: that block edits coefficients on a $\lambda = 10^{-6}$ fit in which the direct boiler term survives in all 20 replicates, so the pathway is never severed by sparsification there, and none of its contrasts reaches significance under either objective.

4.2. A cheap open-loop rule for selecting a control-oriented surrogate

Two things follow for practice, and they are different in kind. The first is a screening rule: score candidates open-loop on (i) free-running multi-step rollout error at a horizon no shorter than the one the model will be used at and (ii) the fraction of rollouts that diverge—both computed on the identification data in seconds, before any closed-loop season is simulated—and keep one-step error out of the gate. The condition number belongs in the report as a diagnostic but not in the ranking: it predicts (i) and (ii) accurately and closed-loop margin not at all (Section 4.1). Ours scored 96 steps (24 h) against a 20-step (5 h) control horizon—a stress test rather than a matched one, so the screening horizon has to be reported with the score. In our ladder the raw configuration dominates the frozen physics-informed recipe on all three pre-registered axes at once (Section 3.1), and the two controllers built on that library then take the top two places in closed loop. The frozen recipe leads on exactly one metric, one-step accuracy (rank 8 of 72 against rank 10), which was not a gate. The screen is close to free: the whole ladder cell costs a few seconds per fit, against a simulated season per candidate for a closed-loop evaluation.

The strength of that claim must be stated at the resolution the three libraries allow. Over the series, the screen *identifies the closed-loop winner* and one-step accuracy ranks that same winner *last*; but neither open-loop criterion reproduces the full closed-loop ordering, because rollout error places full physics third where the closed loop places it second. What is established is the choice of the top candidate and the rejection of one-step accuracy as a selection criterion—not a rank correlation across the series.

The second thing that follows is not a ranking but a check, and it is the one the new mechanism makes necessary. After sparsification, read the fitted coefficient matrix and confirm that every actuator the controller is allowed to move still has a surviving path to the states it is meant to move. The check costs nothing—it inspects numbers already computed—and it is the variable the closed loop turned out to be sensitive to, whereas a well-conditioned regression is not a guarantee that the check passes and an ill-conditioned one is not a guarantee that it fails. When the check does fail, the available responses are a lower threshold, a different basis, or a constraint protecting the actuator columns; we tested only the first, by hand, on one coefficient, and it bought back a median of $+0.21$ EUR m^{-2}.

Two qualifications. The ladder's rollout value is a *comparator*, not a specification: along the oracle controller's own closed-loop trajectory the same frozen recipe predicts indoor temperature to 2.62 °C at 24 h, four times better than the 10.50 °C it scores on PRBS-excited identification data. The held-out block supports the ordering but licenses a different claim: the ordering is identical in-sample and held-out, yet the dominant factor is the *evaluation* year. The physics library without cross terms scores a median rollout error of 16.9 °C evaluated on 2018 and 2.7 °C on 2019 whatever year it was fitted on, and in one of the four fit→evaluate directions it narrowly beats the raw library (2.687 against 2.722). What is defensible is that the raw library is stable regardless of evaluation year while the physics libraries are year-dependent.

The third piece of guidance is therefore negative: do not treat a basis function's physical motivation as evidence that it belongs in the library. Every feature added here is an algebraic function of columns already present, and the two groups in which they were added both cost multi-step stability—the three non-cross features take rollout error from 3.31 to 7.11 °C, the four cross terms take it on to 30.84—while neither group buys back a closed-loop margin above the raw library's. Nor does the *count* of added features order the damage: adding three is worse in closed loop than adding seven (Section 4.1), so a practitioner cannot reason from "I added only a few, physically sensible terms" to "the cost is small". What we cannot say is which individual feature is responsible for anything: they entered in two groups of three and four, no feature was added or removed alone, and the one single-feature deletion that would matter most is the untested experiment named above. Motivation earns a candidate a place in the ladder, and nothing more.

4.3. Sparsity is exonerated

The winner is the sparser model on both counts—11 candidate features against 14, and 19.9 retained terms against 28.25—so sparsity cannot be what the physics-informed variants lost. Nor is more sparsity better within a library: over the sparsity sweep of Table 10 margin falls as terms are removed, and violation steps rise with it, from about 3850 to 9090 between the endpoints. The decline is not monotone across the thirteen levels, and the between-replicate SD reaches 4.14, larger than several of the steps, so only the endpoints carry the comparison. The three-library series settles the question in the other direction as well: across the ensemble triple, in which the library is the only thing that changes, retained terms run $20.2 \rightarrow 28.25 \rightarrow 37.2$ while closed-loop margin runs $+4.32 \rightarrow +0.28 \rightarrow +2.75$ EUR m^{-2}. The least sparse model of the three is not the worst; the middle one is. Term count orders performance in neither direction; the basis does. The two raw-library variants—bagged ensemble and plain sequential thresholded least squares—cannot be separated in closed loop (median $\Delta = -0.01$, mean $+0.48$ EUR m^{-2}, $33/80$ wins, $p = 0.60$), which locates the effect in the library rather than in the estimator. This is a failure to reject, not a demonstration of equivalence: the mean gap is not negligible, only too disperse to establish at $n = 80$.

4.4. What the two comparison defects imply for the field

Both defects (Table 15, Figure 5) were on our side of the comparison and both inflated the result.

The reference heuristic was described as tuned and was not. Searching its four economically relevant setpoints under exactly the budget the reinforcement-learning agents received—16 random trials, selection on the training years only—is worth $+3.49$ EUR m^{-2} and moves the heuristic from last-but-three to seventh of fifteen. The apparent advantage of the raw-library controller over it falls from $+5.54$ to $+2.06\ \mathrm{EUR\,m^{-2}}$: an untuned baseline had inflated the result 2.7-fold. Only 2 of the 16 random draws beat the stock setpoints: the stock configuration sat at the 88th percentile of its own search space—not a straw man, merely un-searched. Tuning also moved the heuristic *onto* the two-axis front, where untuned it had been dominated. Tuning budgets should be reported in units comparable to those granted to the learned controllers [42].

The second defect was internal to the controller. The MPC stage cost weighted boiler, lamp and CO_2 in the ratio $1{:}\,0.5{:}\,0.1$, while the marginal costs the economic criterion actually charges are $1{:}\,2.97{:}\,0.46$: the boiler was penalised twice as hard as the lamp although the lamp costs three times more. Re-weighting the energy term to the measured ratios, its total at $u = (1{,}1{,}1)$ held fixed, improved *both* axes for all eight MPC controllers, but unevenly: the five physics-informed variants gained $+1.54 \ldots + 2.74\ \mathrm{EUR\,m^{-2}}$ and the neural-network MPC $+1.57$, against $+0.21 \ldots + 0.25$ for the two raw ones. Taking the best physics-informed variant, the raw-minus-physics gap fell from $+3.66$ to $+1.23\ \mathrm{EUR\,m^{-2}}$: roughly two-thirds of what had looked like a library effect was a defect of the objective. The repair also cost the tuned heuristic its place on the two-axis front, yielded to a physics-informed variant it had beaten under the defective objective—a baseline's standing in a two-axis comparison is no more stable than the stage cost the controllers were given. Why the correction is worth an order of magnitude more to one library than to the other we did not measure; what follows practically is that a controller defect can masquerade as a property of the model class, and that the stage cost should be checked against the prices in the evaluation criterion before any difference is attributed to a model.

4.5. Relation to prior work

The finding instantiates the identification-for-control programme—model quality cannot be judged independently of the control task the model serves [15–17,22,43]—and the objective-mismatch literature of model-based reinforcement learning, where prediction likelihood is a poor proxy for control performance [18–21]. Our contribution there is constructive rather than diagnostic: an open-loop screen costing seconds per candidate identifies the closed-loop winner, whereas the standard screen ranks that winner *last*. The screen needs only free-running rollouts, so it transfers in principle to any model class; but we exercised it on sparse-regression libraries, and over the three of them it selects the winner without reproducing the ordering of the two losers (Section 4.2), so the correspondence is established as a choice of top candidate, not as a rank correlation and not across model classes. Section 4.4 adds a second, less-studied mismatch: stage-cost weights against the prices in the criterion, inside an otherwise correctly specified problem. Collinearity in the library is what reweighted and balance-guided sparse regression is designed to mitigate [25–26,44], and our open-loop measurements agree with that literature exactly: κ orders multi-step stability without exception. What we do *not* find is the further step from conditioning to closed-loop economics, where the ordering breaks (Section 4.1). The consequence is not that library design matters less but that it has to be scored on what the controller actually uses—a point on which library optimisation scored by recursive long-term prediction [14] converges from the opposite direction. Bagged ensembling [10] changed neither outcome: it is statistically indistinguishable from plain thresholded least squares on the raw library ($p = 0.60$) and on the full physics library ($p = 0.057$). Greenhouse comparisons of MPC and reinforcement learning have generally fixed the model and varied the controller [6,11–13]; the library appears at least as consequential.

4.6. Limitations

- **The advantage is not Pareto-dominant.** Five controllers hold the front; PPO incurs 1330 violation steps against the raw library's 4130, at 3.86 $\mathrm{EUR\,m^{-2}}$ less margin. Superiority is conditional on how a grower weights margin against corridor excursions.
- **The MPC objective is necessarily a proxy.** The surrogate carries no crop state, so revenue cannot be predicted and terminal biomass cannot be priced—structural to the model class, and it blocks the natural terminal-cost formulation.
- **The tuned-heuristic comparison is not a paired test.** The heuristic is deterministic and ran once per season, so its value is a constant reused across replicates and carries no sampling uncertainty: the +2.06 above is a one-sample signed-rank against that constant. Genuinely paired it collapses to one replicate's four seasons (+2.55 $\mathrm{EUR\,m^{-2}}$, $n = 4$), where the minimum attainable p is 0.125 and nothing can be significant.
- **Significance is not magnitude.** Two physics-informed variants differ by 0.0066 $\mathrm{EUR\,m^{-2}}$ over four seasons yet separate at $p = 0.012$, so listing both as front points is an artefact. Effect sizes accompany every p-value, and Holm corrections are over a declared family of 15 controllers—another declaration moves them. This is not hypothetical: adding the two full-physics controllers enlarged the family from 13 to 15 and moved every corrected level in the main comparison, the raw-versus-physics contrast from 7.7×10^{-4} to 4.6×10^{-4} among them, without any controller crossing 0.05.
- **Coverage is uneven.** The neural-network MPC has 10 replicates and 40 runs against 20 and 80 elsewhere, 14 of them terminating early. The oracle has no 2022 season—all 20 runs exhausted the solver-failure budget—but that is an artefact: at a thousandfold budget the season completes at −1.12 and −0.56 $\mathrm{EUR\,m^{-2}}$, so the reported −2.65 is a three-season mean and the imputed four-season value is ≈ −2.20. One front member owes part of its position to truncation: over complete runs its violation margin narrows from 71 to 26 steps.
- **The mechanism rests on a three-point library comparison, which is few.** A claim about what does and does not order closed-loop performance is being carried by three libraries, and the third point exists on one optimiser only: the ensemble triple is complete, whereas at threshold 0.05 under sequential thresholded least squares only the raw and full-physics libraries were run (+3.83 and +2.48 $\mathrm{EUR\,m^{-2}}$), so the raw-over-physics direction replicates but the non-monotonicity itself does not exist as a three-point series on that axis. Boiler-term survival is *measured, not manipulated*, at library level: the libraries differ in the basis, and survival is a downstream property of the resulting fits. The only randomised manipulation anywhere in this argument is the single-coefficient ablation, whose median effect is +0.21 $\mathrm{EUR\,m^{-2}}$—so the mechanism is supported by one controlled contrast of modest size plus a library pattern that is measured rather than manipulated. A fourth library has already been added—the term-deletion probe below—and did not disturb the pattern, but the pattern remains observational at library level.
- **What sets boiler-term survival is unresolved.** The candidate structural explanation—that the bilinear $T_{\mathrm{in}}u_{\mathrm{Boil}}$ term is what rescues the full physics library—has been tested by deleting exactly that feature, same optimiser and threshold, four seasons × 20 replicates in closed loop, and is rejected (Section 4.1): the 17-feature library keeps the full library's conditioning, open-loop instability and closed-loop level, and its survival (0.40) comes nowhere near the middle library's 0.15. The two observational figures once offered for the detour were non-significant at 9–11 and 3–17 replicates and now stand only as absence of contradiction. The non-monotonicity is an established fact without an established cause.
- **The mechanism block is single-season.** All 800 mechanism runs are for 2020, so neither the λ sweep nor the ablation generalises across seasons. The knock-in and cross-term blocks rest on different baselines (boiler-term survival 0.15 against 1.00) and are not mutually comparable; the split by term survival is observational and

confounded with recipe, and only the knock-in contrast is controlled. The library comparison, by contrast, covers all four test seasons, including the term-deletion probe.

- **The held-out check rests on two training years.** With 2018 and 2019 alone, year-dependence cannot be separated from chance, and the evaluation year dominates (Section 4.2).
- **The design and sensitivity blocks were never run under the priced objective.** The support harness hard-codes the default stage cost, so those sensitivities are default-objective results, mislabelled as priced in the run log. Under it, perturbing identified coefficients by $\pm 15\,\%$ and $\pm 20\,\%$ degrades margin significantly (9 of 10 and 10 of 10 replicates worse, span of means $12.2\ \mathrm{EUR\,m^{-2}}$), but as a growing minority of catastrophic runs, not a uniform decline—the median at $\pm 20\,\%$ is only -0.31. This finer grid stops at $\pm 20\,\%$, so the non-monotonicity the coarser canonical grid shows at $\pm 30\,\%$ is unreproduced rather than refuted.
- **The horizon was never tuned for any MPC.** Every MPC improves at horizon 8 over horizon 20, by $+0.29$ for the raw library and $+1.3 \ldots + 2.1$ for the physics-informed variants—consistent with rollout error compounding inside the prediction horizon, which is the open-loop half of the argument and is unaffected by Section 4.1; and an asymmetry running against the MPCs, which received no hyperparameter search while the reinforcement learning agents and the heuristic received 16 trials each.
- **Reproducibility is within-environment only, and cross-environment reproducibility was never measured.** The pipeline reproduces nine digests bit-for-bit on one machine. Whether closed-loop margins survive a change of machine is *unknown*: every wave reported here ran in one environment, and the manifests do not even record which one (Section 2.9). What is measurable in the tree is that closed-loop margins move under changes far smaller than a change of platform—the two harnesses score the *deterministic* heuristic differently in all four seasons at identical configuration hash, by up to $0.038\ \mathrm{EUR\,m^{-2}}$ and monotonically more with each season. A controller with no stochastic element reproducing itself only to two decimal places is the conservative reading of this limitation, and it is enough to warrant the rule adopted: whole waves are regenerated in one environment and never mixed cell-by-cell. The configuration hash does not discriminate between objective variants either, so the results tree and block, not the hash, are the identifying key.
- **Negative results stand as such.** The out-of-distribution guard does not pay for itself, and the sign of its effect depends on which statistic is taken — mean against median, all pairs against completed seasons (Section 3.7). It truncates a left tail rather than raising the centre, costs violations, ends $30\,\%$ of its seasons early, and rests on a detector barely better than chance; its activation count was never recorded, so the natural explanatory variable is missing. The extended Kalman filter observer is worse than doing nothing (-2.95 against $-1.58\ \mathrm{EUR\,m^{-2}}$), $39\,\%$ of its seasons terminating early.
- **Scope.** Simulation only, one location, one planting date, one crop model, one weather source. The ranking is not price-invariant: two different controllers win across a nine-point grid of fruit and energy prices—though that grid re-scores rather than re-optimises and contains neither raw-library controller, so it does not test the ranking reported here. No recursive feasibility or stability guarantee is claimed or proved [4].

Table 14. The selection reversal, measured on the held-out identification block. Rollout RMSE is the median over replicates at the 96-step (24 h) horizon for indoor temperature, with the interquartile range beside it; κ is the condition number of the regression matrix; "terms" counts non-zero coefficients after thresholding; $n = 40$ per cell (10 replicates $\times$ 2 sparse estimators $\times$ 2 fit years). κ and the term count are properties of the fit and so repeat across regimes. The ordering by one-step error is the exact inverse of the ordering by every multi-step criterion, in both regimes—but only over the two sparse estimators used here; pooled over all four estimators of the full ladder the one-step ordering is no longer strict. The medians overstate the separation: both physics libraries are strongly dispersed, and the lower quartile of the no-cross library falls to the raw library's level,

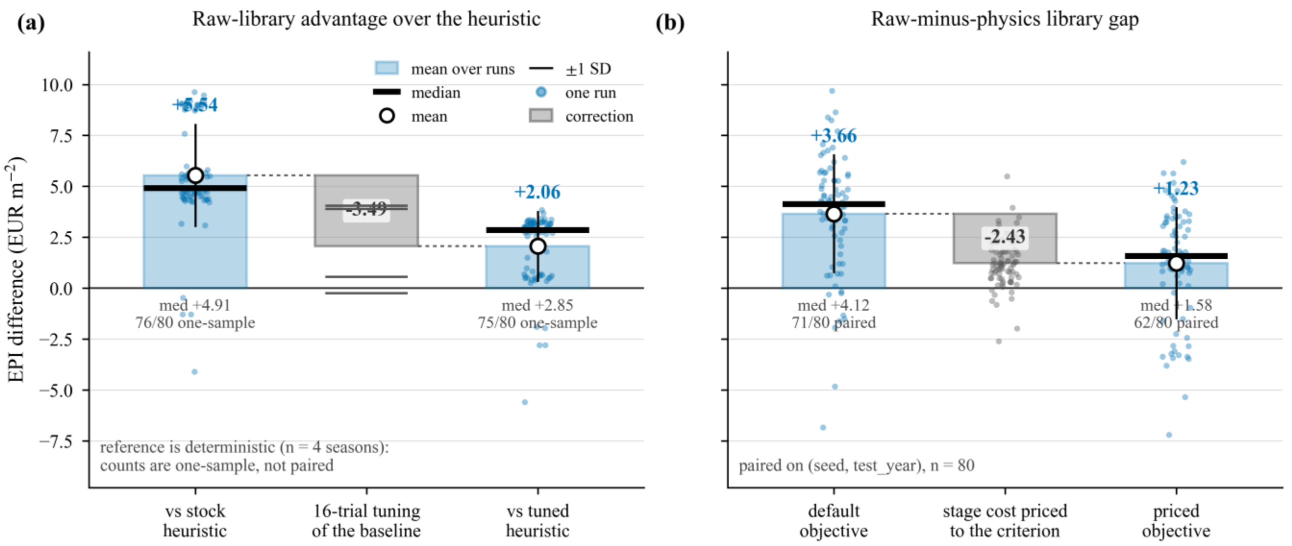
which Section 4.2 attributes to the evaluation year. The table is an *open-loop* measurement throughout: κ orders every column here, and it does not order closed-loop margin, where the series is non-monotone (Section 4.1). The column is reported as a diagnostic and must not be read as a performance ranking.

Feature library	**Features**	κ	**Terms**	**One-step RMSE**	**Rollout RMSE, median [IQR]**	**Diverged**
				(°C)	(°C)	(fraction)
			In-sample			
Raw (physics-free)	11	**8.2**	**20.6**	1.84	**2.62** [2.53, 2.80]	**0.000**
Physics, no cross terms	14	25.4	29.7	1.69	12.14 [2.88, 14.10]	0.021
Physics, full	18	56.2	36.6	**1.64**	21.88 [18.06, 25.58]	0.078
			Held-out year			
Raw (physics-free)	11	**8.2**	**20.6**	1.90	**2.72** [2.63, 2.84]	**0.000**
Physics, no cross terms	14	25.4	29.7	1.82	15.42 [2.69, 21.13]	0.046
Physics, full	18	56.2	36.6	**1.77**	21.57 [15.56, 31.74]	0.086

Table 15. The two comparison defects and what repairing each one cost the headline result. Both were on the authors' side of the comparison and both had inflated it. EPI is the seasonal margin returned by the simulator.

	***Defect 1*: untuned baseline**	***Defect 2*: mis-weighted stage cost**
What was wrong	the reference heuristic was called "tuned" but its set-points were hard-coded and never searched	actuators weighted 1: 0.5: 0.1 (boiler:lamp:CO_2) while the measured marginal costs are 1: 2.97: 0.46

	Defect 1: **untuned baseline**	*Defect 2*: **mis-weighted stage cost**
Repair	16 random trials, the budget the reinforcement-learning agents received; selection on training years only. The MPCs received no search at all, an asymmetry running against them	energy term re-weighted to the measured ratios, its total at $u = (1,1,1)$ held fixed
Direct effect	heuristic $-1.23 \rightarrow +2.26$ EUR m^{-2} (+3.49)	both axes improve for all eight MPC controllers, every contrast Holm-significant, but by $+1.54 \ldots +2.74$ for the physics-informed libraries against $+0.21 \ldots +0.25$ for the raw ones
Effect on the claim	raw-library advantage over the heuristic $+5.54 \rightarrow +2.06$ EUR m^{-2} (2.7 × smaller)	raw-minus-physics gap $+3.66 \rightarrow +1.23$ EUR m^{-2} (two-thirds removed)
Also changed	the heuristic moves onto the two-axis front, which it had not held untuned	it leaves that front again, dominated on both axes; and the boiler-coefficient ablation falls from median $+3.05$ to $+0.21$ EUR m^{-2}



Positive = the raw library is ahead. Panel (a): both ends from n2_tune/tune_rb_n2.csv so the waterfall closes exactly. Panel (b): n7/main_n7.csv and final/main.csv (default) against the priced pool.

Figure 5. Where the apparent effect went. Repairing the untuned baseline removed a factor of 2.7 from the raw library's advantage over the agronomic heuristic; pricing the stage cost consistently with the evaluation criterion removed two-thirds of the apparent library effect. Both corrections leave the direction of the result intact and its magnitude much smaller. Source files are named in the LaTeX comment preceding this caption.

5. Conclusions

We asked which open-loop criterion should select a data-driven surrogate for economic MPC, and what explains its success. Multi-step stability selects; survival of the actuator term through the sparsity threshold, not conditioning, explains (Figure 6).

1. **One-step accuracy and multi-step stability rank the candidate models in opposite orders.** In the first-order, undenoised block of a 72-configuration ladder, under two sparse estimators, physically motivated features lowered one-step RMSE from 1.86 to 1.68 °C but raised the median 24 h rollout RMSE from 2.67 to 24.27 °C. The reversal is confined to that block — pooled over all 72 labels the raw library predicts better one step ahead too — but it reproduces on a held-out year [14,16–18,21].
2. **The library that multi-step stability selects also gives the best closed-loop economics, once both comparison defects are corrected.** Across fifteen controllers on four test seasons the raw-library ensemble surrogate [10] led with $+4.32\ \mathrm{EUR\,m^{-2}}$, against $+3.09$ for the best physics-informed variant (Holm-corrected $p = 4.6 \times 10^{-4}$, family of 15) and $+2.26$ for the agronomic heuristic, re-tuned under the reinforcement-learning agents' own 16-trial budget ($\Delta = +2.06$, 75 of 80 runs against a deterministic reference) [30–31]. Tuning was worth $+3.49$, and repricing the actuator weights cut the raw-minus-physics gap from $+3.66$ to $+1.23$. The direction survives, the magnitude does not.
3. **Physically motivated features hurt, but non-monotonically, and conditioning does not explain the economics.** Changing the feature library alone — same estimator, threshold and seeds — the condition number rises monotonically, $8.2 \to 24.5 \to 53.4$, the fits grow *denser* rather than sparser ($20.1 \to 37.0$ terms), and open-loop rollout error rises with them, $2.67 \to 10.58 \to 24.27$ °C. Closed-loop margin does not: $+4.32 \to +0.28 \to +2.75\ \mathrm{EUR\,m^{-2}}$, the worst-conditioned library beating the intermediate one almost tenfold, with the same shape under the other estimator. Conditioning predicts open-loop stability cleanly and closed-loop economics not at all [25–26,44].
4. **The closed-loop outcome follows whether the actuator pathway survives thresholding.** The direct boiler coefficient survives in 55, 15 and 55 % of fits of the three nested libraries — the ordering of the margins, which conditioning does not reproduce — and in 40 % of fits of the 17-feature deletion library, whose margin ($+2.11\ \mathrm{EUR\,m^{-2}}$) lands between, again in survival order. Reinstating it in the recipe that loses it is significant but small and right-skewed: median $+0.21\ \mathrm{EUR\,m^{-2}}$, mean $+1.92$, 16 of 20 replicates, $p = 5.9 \times 10^{-4}$, one season. Stratifying that recipe's own fits by survival agrees, not significantly ($p = 0.053$). The bilinear temperature–boiler detour once offered as the structural explanation has been tested by deleting that single feature and is rejected; why the middle library loses the term is open, that survival gates the outcome is not.
5. **The advantage is neither Pareto-dominant nor unconditional.** The margin–violation front holds five of the fifteen controllers, and PPO [27] incurs 1330 violation steps against the raw library's 4130 at $3.86\ \mathrm{EUR\,m^{-2}}$ less margin — though two of those points are one controller under two thresholds and a third rests partly on seven truncated runs. The surrogate has no crop state, so terminal biomass is unpriced; the out-of-distribution guard is a negative result (paired mean $-0.69\ \mathrm{EUR\,m^{-2}}$, $+430$ violation steps); a ± 15 % coefficient perturbation costs 7.97 on the mean against 3.35 on the median ($p = 4.3 \times 10^{-3}$); and all of it is simulation, one location, one planting date, one computing environment.

Table 16 lists every headline quantity above together with the result file it is read from, so that each can be checked at source.

Table 16. Headline quantities of the study and the result files they are read from. All paths are relative to own-article/regen/results/. EPI is the seasonal economic performance indicator (margin) returned by the simulator; n counts runs after deduplication on (method, seed, test year). Where three values are given they are the three feature libraries in the order raw / physics-no-cross / physics.

Quantity	Value	Source file
Identification ladder (degree 1, no denoising, sparse estimators, $n = 40$ per library)		
One-step RMSE, T_{in} (°C)	1.86 / 1.73 / 1.68	ladder_rerun/ ladder_rerun*.csv
Median 24 h rollout RMSE (°C)	2.67 / 10.58 / 24.27	as above
Divergence fraction	0.000 / 0.020 / 0.076	as above
Condition number κ	8.21 / 24.52 / 53.43	as above
Non-zero terms	20.1 / 28.2 / 37.0	as above
One-factor library change (ensemble, threshold 0.05, priced objective, four seasons, $n = 80$ per library, zero truncated runs). Only the feature library differs between these three controllers.		
Closed-loop EPI (EUR m^{-2})	+4.32 / +0.28 / +2.75	priced_main/*.csv, phys_lib/ main_physlib.csv
Boiler-term survival[b]	0.55 / 0.15 / 0.55	as above
The same change under STLSQ (EUR m^{-2})	+3.83 / — / +2.48	as above
Four test seasons 2020–2023, fifteen controllers ($n = 80$ unless stated). MPC rows use the priced stage cost; PPO, SAC, the heuristic and the full-model planner have no MPC stage cost and are quoted from the canonical wave.		
Raw-library ensemble MPC (EUR m^{-2}; violation steps)	$+4.32 \pm 4.08$; 4130	priced_main/*.csv
Best physics-informed MPC (EUR m^{-2})	$+3.09 \pm 2.71$	priced_main/*.csv
Raw ensemble vs. that variant	$p_{\text{Holm}} = 4.6 \times 10^{-4}$	recomputed over the family of 15
Tuned heuristic (EUR m^{-2})	$+2.26$ ($n = 4$)[a]	n2_tune/tune_rb_n2.csv
Stock heuristic (EUR m^{-2})	-1.23 ($n = 4$)[a]	n2_tune/tune_rb_n2.csv
PPO (EUR m^{-2}; violation steps)	$+0.45 \pm 2.97$; 1330	final/main.csv
Unseen seasons 2014–2017, default objective ($n = 32$, 8 replicates, four controllers only)		
Raw-library ensemble MPC (EUR m^{-2})	$+6.42 \pm 0.95$	n5_years/main_n5.csv

Quantity	Value	Source file
Untuned heuristic (EUR m^{-2})	-0.33	n5_years/main_n5.csv
Corrections, mechanism and robustness		
Gain from tuning the heuristic (EUR m^{-2})	$+3.49$	n2_tune/tune_rb_n2.csv
Raw-minus-physics gap, default → priced (EUR m^{-2})	$+3.66 \rightarrow +1.23$	priced_main/*.csv, priced_dagger/*.csv, final/main.csv, n7/main_n7.csv
Boiler-coefficient knock-in, priced: median / mean	$+0.21$ / $+1.92$ (16 of 20, $p = 5.9 \times 10^{-4}$)	priced_mech/ mechanism_pricedMech*.csv
$\pm 15\,\%$ coefficient perturbation, priced: mean / median cost (EUR m^{-2})	-7.97 / -3.35 ($p = 4.3 \times 10^{-3}$)	design_priced_real/ design_designPriced.csv

[a] The heuristic is deterministic and was run once per season, so $n = 4$ and no run-to-run dispersion exists; the dispersion of its seasonal values must not be read as an error bar, and any signed-rank test against it is a one-sample test against a constant. Both heuristic rows are quoted from the same harness, n2_tune, so that the tuning gain is internally consistent; the canonical wave returns -1.21 rather than -1.23 for the identical deterministic controller, a 0.02 EUR m^{-2} discrepancy discussed with the reproducibility limits. The two values must not be mixed within one table. [b] Fraction of identification replicates in which the direct boiler-to-temperature coefficient is non-zero after thresholding. Survival is a property of the fit rather than of the run, so its effective n is 20 seeds, not 80 runs.

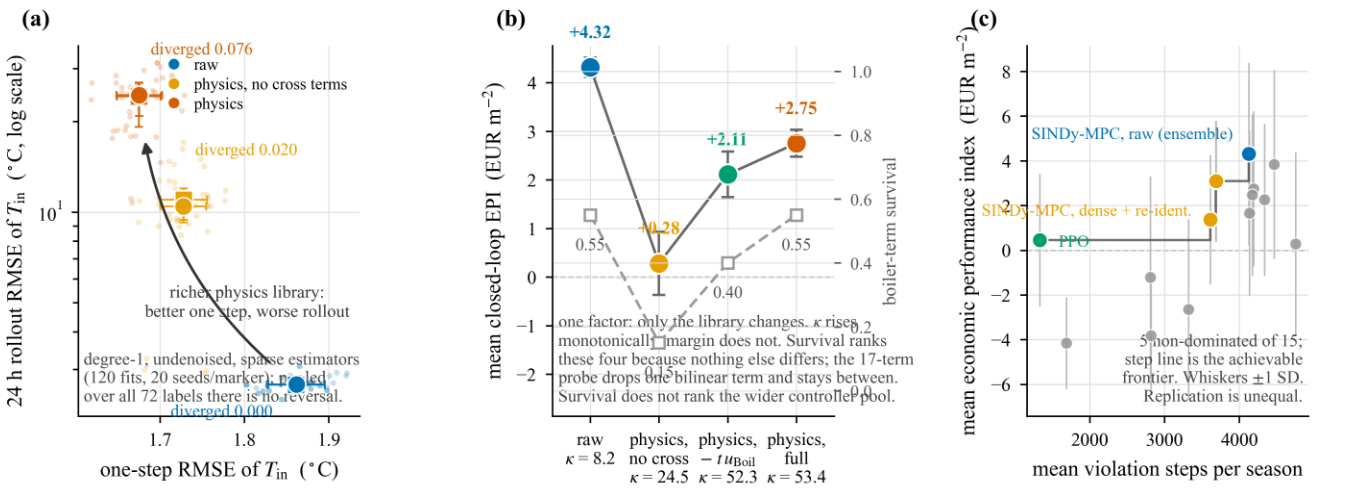


Figure 6. *Graphical abstract.* (**a**) The selection reversal in the first-order, undenoised block under sparse estimators: candidate libraries ordered by one-step prediction error are ordered inversely by 24 h rollout error and by divergence rate. (**b**) Changing only the feature library, the condition number and the open-loop rollout error rise monotonically across the nested libraries while closed-loop margin does not — the worst-conditioned library outperforms the intermediate one, and the 17-feature probe that deletes the single bilinear boiler term stays at the full library's level, rejecting the bypass explanation — and the margin instead follows the fraction of fits in which the direct boiler coefficient survives thresholding. (**c**) The two-axis outcome over four test seasons under the priced objective: the front holds five of the fifteen controllers, so the economic advantage of the raw library is not Pareto-dominant. Sources are named in the LaTeX comment preceding this caption.

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Author Contributions: [[REQUIRED — NOT SUPPLIED.]] State each author's contribution by initials using the CRediT taxonomy, e.g. "Conceptualization, X.X. and Y.Y.; methodology, X.X.; software, X.X.; validation, X.X., Y.Y. and Z.Z.; formal analysis, X.X.; investigation, X.X.; resources, X.X.; data curation, X.X.; writing—original draft preparation, X.X.; writing—review and editing, X.X.; visualization, X.X.; supervision, X.X.; project administration, X.X.; funding acquisition, Y.Y. All authors have read and agreed to the published version of the manuscript." Initials must match the author block above, and every listed author must appear in at least one role.

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Institutional Review Board Statement: Not applicable. This study is a computational study of a greenhouse climate simulator and involves neither humans nor animals.

Informed Consent Statement: Not applicable.

Data Availability Statement: All quantities reported in this article are computed from the per-run result tables of the study's regeneration tree (own-article/regen/results/ in the project repository), which contains one row per (controller, seed, test season) together with the seasonal margin, constraint-violation counts, solver diagnostics and the identified sparse coefficients. Every wave reported here was produced under a single frozen configuration whose hash, 637c6b535a9e, is written into every result row and into the regen_manifest.json of each wave alongside the git commit of the code that generated it; the manifest also records the seeds, test years, horizon, solver-failure budget, season length and the four identification recipes, so a wave can be identified without reference to this text. The experiments are regenerated with the single driver in own-article/regen/: each of the eight experiment blocks is run as python run_regen.py --experiment <block> --seeds <list> --out <dir>, after which python run_regen.py --merge --out <dir> and python make_tables.py --out <dir> rebuild every derived table together with NUMBERS.md, a claim-to-value-to-source map that names the file and column behind each reported quantity. python verify_regen.py then applies the acceptance gates—mixed configuration hash, an incomplete or duplicated run grid, truncated seasons, non-uniform horizon or solver budget, a missing sparse coefficient column, an unresolved sparsity region or an incomplete ablation block—and exits non-zero if any fails. Determinism is checked separately by python repro.py --selftest, which executes the pipeline twice under identical seeds and compares SHA-256 digests of the training states and actions, the ensemble and STLSQ coefficient matrices, the neural-network weights, and the closed-loop trajectory and margin; all seven digests agree on the pinned stack, and all nine with the two reinforcement-learning policies included. Reproduction additionally requires the GreenLight tomato greenhouse model as packaged in gl_gym 0.3.1 [1,13] and the ERA5-derived weather described in Section 2.1, retrieved from the Open-Meteo historical archive API [33–34]. As Section 2.9 states, bit-level reproduction is established within one computing environment; the cross-environment case was not measured, and closed-loop margins should not be expected to match to the last decimal on a different stack. [[REQUIRED BEFORE SUBMISSION: a public, citable location for this tree — a repository URL and an archived release with a DOI (Zenodo, figshare or equivalent). MDPI requires a link or an explicit statement of restriction; a path inside a private repository is not sufficient.]]

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Conflicts of Interest: [[REQUIRED — NOT SUPPLIED.]] Every author must declare. Use "The authors declare no conflicts of interest." only if that is true of all of them; otherwise describe the interest and state the funders' role, or lack of role, in the design of the study, in the collection, analyses or interpretation of data, in the writing of the manuscript, and in the decision to publish the results.

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